Advanced Econometrics — MPE (Insper)

Problem set: *L03 — Introduction to Bayesian learning (Classes 3+4): textbook exercises*
Solutions

Instructions. Every question is a textbook exercise from the "Practice" list of Classes 3+4 in `Map/class-plan.md`, reproduced verbatim from the book PDF with its source line; Portuguese-language books (Bolfarine & Sandoval; Morettin & Singer) are quoted in Portuguese. When an exercise refers to an equation, table or exercise elsewhere in the book, that material is reproduced as *context*. Show every step. Computational items are solved in Python (NumPy/SciPy only, fixed seeds); every number printed in the solutions is reproduced and asserted by `Listas/L03-book-exercises-check.py`. Greenberg's gamma $G(\alpha,\beta)$ and Bolfarine's $Gama(\alpha,\beta)$ both use the rate form $\beta^{\alpha}\theta^{\alpha-1}e^{-\beta\theta}/\Gamma(\alpha)$, mean α/β; $IG(\alpha,\beta)$ is the inverse gamma with density $\beta^{\alpha}y^{-(\alpha+1)}e^{-\beta/y}/\Gamma(\alpha)$ (Greenberg, App. A.1.7–A.1.8).

Block 1 — Greenberg (2007) *(Chs. 2–8 • 15 questions)*

Question 1. Greenberg 2.2 — gamma–Poisson conjugacy.
Source: Greenberg, E. (2007), *Introduction to Bayesian Econometrics*, Cambridge University Press, Exercise 2.2, p. 19 — reproduced verbatim.

> **2.2** The Poisson distribution has probability mass function
>
> $$p(y_i|\theta) = \frac{\theta^{y_i}e^{-\theta}}{y_i!}, \qquad \theta > 0,\ y_i = 0, 1, \ldots,$$
>
> and let $y_1, \ldots, y_n$ be a random sample from this distribution.
>
> **(a)** Show that the gamma distribution $G(\alpha, \beta)$ is a conjugate prior distribution for the Poisson distribution.
>
> **(b)** Show that $\bar{y}$ is the MLE for θ.
>
> **(c)** Write the mean of the posterior distribution as a weighted average of the mean of the prior distribution and the MLE.
>
> **(d)** What happens to the weight on the prior mean as n becomes large?

Notation.

- $y_i \in \{0, 1, \ldots\}$: count of observation i; $n \geq 1$: sample size; $s = \sum_{i=1}^{n} y_i \geq 0$; $\bar{y} = s/n$.
- $\theta > 0$: Poisson rate (mean); $\alpha, \beta > 0$: prior shape and rate, $G(\alpha,\beta)$ density $\beta^{\alpha}\theta^{\alpha-1}e^{-\beta\theta}/\Gamma(\alpha)$, mean α/β (Greenberg A.1).
- $w \in (0, 1)$: weight on the prior mean.

Step — (a) likelihood. By independence the joint p.m.f. is the product of the statement's p.m.f.:

$$f(y|\theta) = \prod_{i=1}^{n} \frac{\theta^{y_i}e^{-\theta}}{y_i!} \tag{Q1.1}$$

$$= \frac{\theta^{s}e^{-n\theta}}{\prod_i y_i!}. \tag{Q1.2}$$

Step — (a) prior and Bayes theorem. The prior and Bayes theorem are

$$\pi(\theta) = \frac{\beta^{\alpha}}{\Gamma(\alpha)}\theta^{\alpha-1}e^{-\beta\theta}, \tag{Q1.3}$$

$$\pi(\theta|y) \propto f(y|\theta)\,\pi(\theta). \tag{Q1.4}$$

Substituting (Q1.2) and (Q1.3) into (Q1.4) and dropping factors free of θ:

$$\pi(\theta|y) \propto \left(\theta^{s}e^{-n\theta}\right)\left(\theta^{\alpha-1}e^{-\beta\theta}\right) \tag{Q1.5}$$

$$= \theta^{\alpha+s-1} e^{-(\beta+n)\theta}. \tag{Q1.6}$$

(Q1.6) is the kernel of $G(\alpha+s, \beta+n)$: prior and posterior are in the same family, so $G(\alpha,\beta)$ is conjugate.

Step — (b) MLE. The log of (Q1.2) and its derivative are

$$\ell(\theta) = s\log\theta - n\theta - \textstyle\sum_i \log y_i!, \tag{Q1.7}$$

$$\ell'(\theta) = \frac{s}{\theta} - n. \tag{Q1.8}$$

Setting (Q1.8) to zero gives $\hat\theta = s/n = \bar y$; $\ell''(\theta) = -s/\theta^2 < 0$ for $s > 0$, so it is a maximum (if $s = 0$, (Q1.7) decreases in θ and the supremum is at $\bar y = 0$).

Step — (c) posterior mean. From (Q1.6) and the gamma mean,

$$E(\theta|y) = \frac{\alpha+s}{\beta+n}. \tag{Q1.9}$$

Writing $\alpha = \beta(\alpha/\beta)$ and $s = n\bar y$ in the numerator of (Q1.9):

$$E(\theta|y) = \frac{\beta\,(\alpha/\beta) + n\,(\bar y)}{\beta+n} \tag{Q1.10}$$

$$= \underbrace{\frac{\beta}{\beta+n}}_{w}\,\frac{\alpha}{\beta} + \underbrace{\frac{n}{\beta+n}}_{1-w}\,\bar y. \tag{Q1.11}$$

Step — (d) large n. $w = \beta/(\beta+n) \to 0$ as $n \to \infty$, so by (Q1.11) $E(\theta|y) \to \bar y$: the data swamp the prior.

Step — numerical illustration. Coal-mining disasters 1851–1891 (the series of the class-1 Rmd page): $n = 41$, $s = 127$, $\bar y = 127/41 = 3.097561$. With $G(2,1)$, (Q1.6) gives $G(129, 42)$; (Q1.9) gives $129/42 = 3.071429$; $w = 1/(1+41) = 0.023810$, and (Q1.11) reads $(0.023810)(2) + (0.976190)(3.097561) = 0.047619 + 3.023810 = 3.071429$.

Intuition. β acts as a prior "number of observations" and α as the prior "total count": the posterior pools β pseudo-observations with the n real ones, so the prior's weight is its share of the pooled sample.

Ref: Greenberg §2.2 (pp. 12–18) and App. A.1.7; Tsay Result 12.4 (p. 620); Bolfarine & Sandoval Example 4.4.2 (pp. 67–68).

Question 2. Greenberg 2.4 — Pareto–uniform conjugacy.
Source: Greenberg, E. (2007), *Introduction to Bayesian Econometrics*, Cambridge University Press, Exercise 2.4, p. 19 — reproduced verbatim.

> **2.4** Consider the uniform distribution with density function $f(y_i|\theta) = 1/\theta$, $0 \le y_i \le \theta$, and θ unknown.
>
> **(a)** Show that the Pareto distribution,
>
> $$\pi(\theta) = \begin{cases} a k^a \theta^{-(a+1)}, & \theta \ge k,\ a > 0, \\ 0, & \text{otherwise}, \end{cases}$$
>
> is a conjugate prior distribution for the uniform distribution.
>
> **(b)** Show that $\hat\theta = \max(y_1, \ldots, y_n)$ is the MLE of θ, where the y_i are a random sample from $f(y_i|\theta)$.
>
> **(c)** Find the posterior distribution of θ and its expected value.

Notation.

- $y_i \geq 0$: observations, $i = 1, \ldots, n$; $y_{(n)} = \max_i y_i$; $\theta > 0$: upper end of the support.
- $k > 0$: Pareto scale (lower bound of θ); $a > 0$: Pareto shape; $1(\cdot)$: indicator.
- $k_1 = \max(k, y_{(n)})$, $a_1 = a + n$: posterior scale and shape.

Step — (a) likelihood. Each density is $\theta^{-1}1(y_i \leq \theta)$, so

$$f(y|\theta) = \prod_{i=1}^{n} \theta^{-1} 1(y_i \leq \theta) \tag{Q2.1}$$

$$= \theta^{-n}\, 1(\theta \geq y_{(n)}), \tag{Q2.2}$$

because all $y_i \leq \theta$ exactly when the largest one is.

Step — (a) posterior. With the prior $\pi(\theta) = ak^a\theta^{-(a+1)}1(\theta \geq k)$, substituting (Q2.2) into $\pi(\theta|y) \propto f(y|\theta)\pi(\theta)$:

$$\pi(\theta|y) \propto \left(\theta^{-n}1(\theta \geq y_{(n)})\right)\left(ak^a\theta^{-(a+1)}1(\theta \geq k)\right) \tag{Q2.3}$$

$$\propto \theta^{-(a+n+1)}\, 1(\theta \geq \max(k, y_{(n)})). \tag{Q2.4}$$

The normalising integral of (Q2.4) is

$$\int_{k_1}^{\infty} \theta^{-(a_1+1)} d\theta = \frac{k_1^{-a_1}}{a_1}, \tag{Q2.5}$$

so dividing (Q2.4) by (Q2.5):

$$\pi(\theta|y) = a_1 k_1^{a_1}\theta^{-(a_1+1)}1(\theta \geq k_1), \qquad a_1 = a + n,\ k_1 = \max(k, y_{(n)}). \tag{Q2.6}$$

(Q2.6) is again Pareto: the family is conjugate.

Step — (b) MLE. By (Q2.2) the likelihood is 0 for $\theta < y_{(n)}$ and θ^{-n} for $\theta \geq y_{(n)}$, where $\frac{d}{d\theta}\theta^{-n} = -n\theta^{-n-1} < 0$. A function that is zero to the left of $y_{(n)}$ and strictly decreasing to its right is maximised at the boundary: $\hat{\theta} = y_{(n)}$ (no derivative equals zero; the maximum is at a corner).

Step — (c) posterior mean. For a Pareto(a_1, k_1) variable with $a_1 > 1$,

$$E(\theta|y) = \int_{k_1}^{\infty} \theta\, a_1 k_1^{a_1}\theta^{-(a_1+1)} d\theta \tag{Q2.7}$$

$$= a_1 k_1^{a_1}\, \frac{k_1^{1-a_1}}{a_1 - 1} \tag{Q2.8}$$

$$= \frac{a_1 k_1}{a_1 - 1}. \tag{Q2.9}$$

Substituting $a_1 = a + n$ and $k_1 = \max(k, y_{(n)})$ into (Q2.9):

$$E(\theta|y) = \frac{(a+n)\max(k, y_{(n)})}{a + n - 1}, \tag{Q2.10}$$

finite because $a + n > 1$ whenever $n \geq 1$.

Intuition. The data can only push the lower bound of θ up (to $y_{(n)}$); the posterior mean sits a factor $(a+n)/(a+n-1) > 1$ above that bound, correcting the downward bias of $y_{(n)}$, and the factor tends to one as n grows.

Ref: Greenberg §2.2; Bolfarine & Sandoval Exercise 4.10 (Question 22 below, same model in other letters) and Example 5.5.2 (p. 86).

Question 3. Greenberg 3.3 — normal predictive distribution.
Source: Greenberg, E. (2007), *Introduction to Bayesian Econometrics*, Cambridge University Press, Exercise 3.3, p. 39 — reproduced verbatim.

> **3.3** Compute the predictive distribution for y_{n+1} if the y_i have independent normal distributions $N(\mu, 1)$, where the prior distribution for μ is $N(\mu_0, \sigma_0^2)$.

Notation.
- $y_i|\mu \sim N(\mu, 1)$, $i = 1, \ldots, n$, independent; $\bar{y}$: sample mean; y_{n+1}: the next observation.
- $\mu \in \mathbb{R}$: unknown mean; $\mu_0 \in \mathbb{R}$, $\sigma_0^2 > 0$: prior mean and variance.
- μ_n, σ_n^2: posterior mean and variance of μ.

Step — posterior of μ. Bayes theorem with the normal likelihood and prior:

$$\pi(\mu|y) \propto \exp\Big[-\tfrac{1}{2}\textstyle\sum_i (y_i-\mu)^2\Big] \exp\Big[-\dfrac{(\mu-\mu_0)^2}{2\sigma_0^2}\Big]. \tag{Q3.1}$$

Expanding the exponent of (Q3.1) and dropping terms free of μ:

$$-\tfrac{1}{2}\Big[n\mu^2 - 2\mu n\bar{y} + \frac{\mu^2}{\sigma_0^2} - \frac{2\mu\mu_0}{\sigma_0^2}\Big] = -\tfrac{1}{2}\Big[\Big(n+\frac{1}{\sigma_0^2}\Big)\mu^2 - 2\mu\Big(n\bar{y}+\frac{\mu_0}{\sigma_0^2}\Big)\Big]. \tag{Q3.2}$$

Define

$$\sigma_n^2 = \Big(n + \frac{1}{\sigma_0^2}\Big)^{-1}, \tag{Q3.3}$$

$$\mu_n = \sigma_n^2\Big(n\bar{y} + \frac{\mu_0}{\sigma_0^2}\Big). \tag{Q3.4}$$

Substituting (Q3.3)–(Q3.4) into (Q3.2) and completing the square:

$$-\frac{1}{2\sigma_n^2}\big[\mu^2 - 2\mu\mu_n\big] = -\frac{(\mu-\mu_n)^2}{2\sigma_n^2} + \text{const}, \tag{Q3.5}$$

so $\mu|y \sim N(\mu_n, \sigma_n^2)$.

Step — predictive. By definition,

$$f(y_{n+1}|y) = \int f(y_{n+1}|\mu)\,\pi(\mu|y)\,d\mu. \tag{Q3.6}$$

Equivalently, write $y_{n+1} = \mu + \varepsilon$ with $\varepsilon \sim N(0,1)$ independent of μ given y. A sum of independent normals is normal, with means and variances adding:

$$E(y_{n+1}|y) = \mu_n + 0 = \mu_n, \tag{Q3.7}$$

$$\mathrm{Var}(y_{n+1}|y) = \sigma_n^2 + 1. \tag{Q3.8}$$

Substituting (Q3.3)–(Q3.4):

$$y_{n+1}|y \sim N\left(\frac{n\bar{y} + \mu_0/\sigma_0^2}{n + 1/\sigma_0^2},\ 1 + \frac{1}{n+1/\sigma_0^2}\right). \tag{Q3.9}$$

The check script evaluates (Q3.6) by quadrature and matches (Q3.9) to 10^{-10}.

Intuition. Two sources of uncertainty add: the sampling noise of a new draw (variance 1) and the remaining uncertainty about μ (σ_n^2). As $n \to \infty$, $\sigma_n^2 \to 0$ and the predictive tends to the "plug-in" $N(\bar{y}, 1)$.

Ref: Greenberg §3.2.3 (prediction) and §2.2; Tsay Result 12.1 (p. 619).

Question 4. Greenberg 3.4 — sequential updating, Poisson model.
Source: Greenberg, E. (2007), *Introduction to Bayesian Econometrics*, Cambridge University Press, Exercise 3.4, p. 39 — reproduced verbatim.

> **3.4** Explicitly verify the updating feature of the posterior distribution for the case where y has a Poisson distribution with parameter θ. Choose $G(\alpha, \beta)$ as the prior distribution, and consider observing a total of s_1 events in the first of two independent experiments and s_2 in the second.

Notation.
- Experiment $j = 1, 2$: $n_j \geq 1$ independent Poisson(θ) counts with total s_j (the statement's case, one count per experiment, is $n_1 = n_2 = 1$).
- $y^{(j)}$: data of experiment j; $\theta > 0$; $\alpha, \beta > 0$: prior parameters.

Step — after experiment 1. As in (Q1.6) of Question 1, with n_1 counts totalling s_1:

$$\pi(\theta|y^{(1)}) \propto \left(\theta^{s_1} e^{-n_1\theta}\right)\left(\theta^{\alpha-1} e^{-\beta\theta}\right) \tag{Q4.1}$$

$$= \theta^{\alpha+s_1-1} e^{-(\beta+n_1)\theta}, \tag{Q4.2}$$

i.e. $G(\alpha + s_1, \beta + n_1)$.

Step — experiment 2 with (Q4.2) **as the prior.** Independence gives $f(y^{(1)}, y^{(2)}|\theta) = f(y^{(1)}|\theta) f(y^{(2)}|\theta)$, so

$$\pi(\theta|y^{(1)}, y^{(2)}) \propto f(y^{(2)}|\theta)\, \pi(\theta|y^{(1)}). \tag{Q4.3}$$

Substituting $f(y^{(2)}|\theta) \propto \theta^{s_2} e^{-n_2\theta}$ and (Q4.2) into (Q4.3):

$$\pi(\theta|y^{(1)}, y^{(2)}) \propto \left(\theta^{s_2} e^{-n_2\theta}\right)\left(\theta^{\alpha+s_1-1} e^{-(\beta+n_1)\theta}\right) \tag{Q4.4}$$

$$= \theta^{\alpha+s_1+s_2-1} e^{-(\beta+n_1+n_2)\theta}. \tag{Q4.5}$$

Step — both experiments at once. The pooled likelihood is $\theta^{s_1+s_2} e^{-(n_1+n_2)\theta}$; times the prior,

$$\pi(\theta|y^{(1)}, y^{(2)}) \propto \left(\theta^{s_1+s_2} e^{-(n_1+n_2)\theta}\right)\left(\theta^{\alpha-1} e^{-\beta\theta}\right) = \theta^{\alpha+s_1+s_2-1} e^{-(\beta+n_1+n_2)\theta}, \tag{Q4.6}$$

identical to (Q4.5): $G(\alpha + s_1 + s_2,\ \beta + n_1 + n_2)$, which is $G(\alpha + s_1 + s_2, \beta + 2)$ for one count per experiment.

Intuition. Yesterday's posterior is today's prior. Because the sufficient statistics (event totals and exposure) simply add, the order and grouping of the data do not matter.

Ref: Greenberg §3.1.3 (updating), pp. 25–26; Tsay Result 12.4.

Question 5. Greenberg 3.2 — Poisson–gamma predictive distribution.
Source: Greenberg, E. (2007), *Introduction to Bayesian Econometrics*, Cambridge University Press, Exercise 3.2, p. 39 — reproduced verbatim.

> **3.2** Compute the predictive distribution for y_{n+1} if the y_i have independent Poisson distributions with parameter θ, given that the first n experiments yielded $y_1, y_2, \ldots, y_n$ events, respectively. Assume the prior distribution $G(\alpha, \beta)$.

Notation.
- y_i, $s = \sum_{i \leq n} y_i$, θ, α, β as in Question 1; $y \in \{0, 1, \ldots\}$: value of y_{n+1}.
- $\alpha_1 = \alpha + s$, $\beta_1 = \beta + n$: posterior parameters.

Step — posterior. By (Q1.6) of Question 1, $\theta|y_{1:n} \sim G(\alpha_1, \beta_1)$.

Step — predictive integral. Integrating the Poisson p.m.f. against the posterior:

$$p(y_{n+1} = y|y_{1:n}) = \int_0^\infty \frac{\theta^y e^{-\theta}}{y!} \frac{\beta_1^{\alpha_1}}{\Gamma(\alpha_1)} \theta^{\alpha_1 - 1} e^{-\beta_1 \theta}\, d\theta \tag{Q5.1}$$

$$= \frac{\beta_1^{\alpha_1}}{\Gamma(\alpha_1)\, y!} \int_0^\infty \theta^{\alpha_1 + y - 1} e^{-(\beta_1 + 1)\theta} d\theta. \tag{Q5.2}$$

The gamma integral (Greenberg A.1.7) is

$$\int_0^\infty \theta^{a-1} e^{-b\theta} d\theta = \frac{\Gamma(a)}{b^a}. \tag{Q5.3}$$

Substituting (Q5.3) with $a = \alpha_1 + y$, $b = \beta_1 + 1$ into (Q5.2):

$$p(y_{n+1} = y|y_{1:n}) = \frac{\beta_1^{\alpha_1}}{\Gamma(\alpha_1)\, y!} \cdot \frac{\Gamma(\alpha_1 + y)}{(\beta_1 + 1)^{\alpha_1 + y}} \tag{Q5.4}$$

$$= \frac{\Gamma(\alpha_1 + y)}{\Gamma(\alpha_1)\, y!} \left(\frac{\beta_1}{\beta_1 + 1}\right)^{\alpha_1} \left(\frac{1}{\beta_1 + 1}\right)^{y}, \qquad y = 0, 1, \ldots \tag{Q5.5}$$

This is the negative binomial with $r = \alpha_1$ and success probability $p = \beta_1/(\beta_1 + 1)$.

Step — moments. For that negative binomial, $E = r(1-p)/p$ and $\mathrm{Var} = r(1-p)/p^2$; substituting $r = \alpha_1$, $p = \beta_1/(\beta_1 + 1)$:

$$E(y_{n+1}|y_{1:n}) = \frac{\alpha_1}{\beta_1}, \tag{Q5.6}$$

$$\mathrm{Var}(y_{n+1}|y_{1:n}) = \frac{\alpha_1}{\beta_1}\Big(1 + \frac{1}{\beta_1}\Big). \tag{Q5.7}$$

Step — numbers. Coal data 1851–1891 with $G(2, 1)$ (Question 1): $\alpha_1 = 129$, $\beta_1 = 42$. From (Q5.5), $P(y_{n+1} = 0) = (42/43)^{129} = 0.048054$ and $P(y_{n+1} = 3) = \frac{131 \cdot 130 \cdot 129}{3!}(42/43)^{129}(1/43)^3 = 0.221297$; from (Q5.6)–(Q5.7), $E = 129/42 = 3.071429$ and $\mathrm{Var} = (3.071429)(1 + 1/42) = 3.144558$. The script also checks (Q5.5) against direct numerical integration of (Q5.1).

Intuition. The predictive variance exceeds the mean (a Poisson with known θ would have them equal): the factor $1 + 1/\beta_1$ is the extra spread coming from not knowing θ, and it vanishes as n (hence β_1) grows.

Ref: Greenberg §3.2.3, App. A.1.3 (negative binomial) and A.1.7.

Question 6. Greenberg 3.8 — Bayes factor for two typists.

Source: Greenberg, E. (2007), *Introduction to Bayesian Econometrics*, Cambridge University Press, Exercise 3.8, p. 40 — reproduced verbatim.

> **3.8** Suppose the number of typographical errors per page has a Poisson distribution $P(\theta)$. Sam types the first m pages of a manuscript and makes e_1 errors in total, and Levi types the last m pages and makes e_2 errors. Let Sam's error rate be θ_1 and Levi's error rate be θ_2. To compare model $M_1 : \theta_1 = \theta_2 = \theta$ with model $M_2 : \theta_1 \neq \theta_2$, produce a table like Table 3.2 for $m = 100$ with 10 total errors and for $m = 200$ with 20 total errors and Sam's error proportions of 0.9, 0.8, 0.7, 0.6, and 0.5. Take $G(1, 1)$ as a prior for θ_1 and θ_2.
>
> *Context the item refers to (not part of the exercise; Greenberg, Table 3.2, p. 38, reproduced verbatim).*
>
> **Table 3.2.** Bayes Factors for Selected Possible Outcomes

Michaela Proportion heads	Lila Proportion heads	$\log_{10}$(Bayes factor) $m=10$	$m=100$
0.1	0.9	−2.506	−30.775
0.2	0.8	−1.200	−15.793
0.3	0.7	−0.348	−6.316
0.4	0.6	0.138	−0.975
0.5	0.5	0.297	0.756

Notation.

- y_{1i}, y_{2i} $(i=1,\dots,m)$: errors on page i typed by Sam and by Levi; $e_1=\sum_i y_{1i}$, $e_2=\sum_i y_{2i}$, $e=e_1+e_2$.
- $\theta,\theta_1,\theta_2>0$: error rates per page; prior $G(1,1)$, i.e. $\pi(\theta)=e^{-\theta}$.
- $P=\prod_i y_{1i}!\prod_i y_{2i}!$; $m_j(y)$: marginal likelihood of M_j; $B_{12}=m_1(y)/m_2(y)$.

Step — marginal likelihood of M_1. All $2m$ pages share θ:

$$m_1(y)=\int_0^\infty \prod_i \frac{\theta^{y_{1i}}e^{-\theta}}{y_{1i}!}\prod_i \frac{\theta^{y_{2i}}e^{-\theta}}{y_{2i}!}\,e^{-\theta}\,d\theta \tag{Q6.1}$$

$$=\frac{1}{P}\int_0^\infty \theta^{e}e^{-(2m+1)\theta}d\theta. \tag{Q6.2}$$

Substituting (Q5.3) (Question 5) with $a=e+1$, $b=2m+1$:

$$m_1(y)=\frac{1}{P}\,\frac{\Gamma(e+1)}{(2m+1)^{e+1}}. \tag{Q6.3}$$

Step — marginal likelihood of M_2. The two rates have independent priors, so the integral factorises:

$$m_2(y)=\frac{1}{P}\int_0^\infty \theta_1^{e_1}e^{-(m+1)\theta_1}d\theta_1\int_0^\infty \theta_2^{e_2}e^{-(m+1)\theta_2}d\theta_2 \tag{Q6.4}$$

$$=\frac{1}{P}\,\frac{\Gamma(e_1+1)}{(m+1)^{e_1+1}}\,\frac{\Gamma(e_2+1)}{(m+1)^{e_2+1}}. \tag{Q6.5}$$

Step — Bayes factor. Dividing (Q6.3) by (Q6.5); P cancels, so only the totals matter:

$$B_{12}=\frac{\Gamma(e+1)\,(m+1)^{e+2}}{(2m+1)^{e+1}\,\Gamma(e_1+1)\,\Gamma(e_2+1)}, \tag{Q6.6}$$

$$\log_{10}B_{12}=\frac{\ln\Gamma(e+1)+(e+2)\ln(m+1)-(e+1)\ln(2m+1)-\ln\Gamma(e_1+1)-\ln\Gamma(e_2+1)}{\ln 10}. \tag{Q6.7}$$

Step — one entry worked out. $m=100$, $e=10$, Sam's proportion 0.9, so $e_1=9$, $e_2=1$. The pieces of (Q6.7): $\ln\Gamma(11)=\ln 10!=15.104413$; $12\ln 101=12(4.615121)=55.381446$; $11\ln 201=11(5.303305)=58.336354$; $\ln\Gamma(10)=\ln 9!=12.801827$; $\ln\Gamma(2)=0$. Substituting:

$$\log_{10}B_{12}=\frac{(15.104413)+(55.381446)-(58.336354)-(12.801827)-(0)}{2.302585} \tag{Q6.8}$$

$$=\frac{-0.652323}{2.302585}=-0.283. \tag{Q6.9}$$

Step — the table. The same computation for every cell (code in the check script, function `g38`; each cell is also verified by integrating (Q6.2) and (Q6.4) numerically):

	$m = 100$, $e = 10$		$m = 200$, $e = 20$	
Sam's proportion	(e_1, e_2)	$\log_{10} B_{12}$	(e_1, e_2)	$\log_{10} B_{12}$
0.9	(9, 1)	−0.283	(18, 2)	−1.717
0.8	(8, 2)	0.370	(16, 4)	−0.310
0.7	(7, 3)	0.796	(14, 6)	0.593
0.6	(6, 4)	1.039	(12, 8)	1.105
0.5	(5, 5)	1.118	(10, 10)	1.271

Step — reading it. On Jeffreys' scale (Greenberg Table 3.1), with 10 errors even a 9–1 split favours M_2 only weakly ($\log_{10} B_{21} = 0.283 < 1/2$), while an even split gives strong support to M_1 ($1 < 1.118 < 3/2$). Doubling the evidence (20 errors) sharpens everything: an 18–2 split now gives very strong evidence for M_2 ($3/2 < 1.717 < 2$), and equal shares strengthen M_1 (1.271).

Intuition. The Bayes factor automatically penalises the extra parameter of M_2: it pays off only when the split is lopsided enough to beat that Occam penalty, and the same proportions carry more weight when they rest on more errors (compare the two m columns, as in Table 3.2).

Ref: Greenberg §3.2.4 (Bayes factors, Table 3.1, Table 3.2), pp. 33–38.

Question 7. Greenberg 4.5 — marginal likelihood of the conjugate normal regression.
Source: Greenberg, E. (2007), *Introduction to Bayesian Econometrics*, Cambridge University Press, Exercise 4.5, p. 58 — reproduced verbatim.

4.5 Since the posterior distribution of (β, σ^2) in Section 4.3 is of known form, it is possible to compute its marginal likelihood analytically. Start from the definition of the marginal likelihood:

$$
\begin{aligned}
&\int f(y|\beta,\sigma^2)\pi(\beta,\sigma^2)\,d\beta\,d\sigma^2 \\
&= \int \left(\frac{1}{2\pi\sigma^2}\right)^{n/2} \exp\left[-\frac{1}{2\sigma^2}(y-X\beta)'(y-X\beta)\right] \left(\frac{1}{2\pi\sigma^2}\right)^{K/2} \\
&\quad \times \frac{1}{|B_0|^{1/2}} \exp\left[-\frac{1}{2\sigma^2}(\beta-\beta_0)'B_0^{-1}(\beta-\beta_0)\right] \\
&\quad \times \frac{(\delta_0/2)^{\alpha_0/2}}{\Gamma(\alpha_0/2)} \left(\frac{1}{\sigma^2}\right)^{\alpha_0/2+1} \exp\left[-\frac{\delta_0}{2\sigma^2}\right] d\beta\,d\sigma^2 \\
&= \left(\frac{1}{2\pi}\right)^{n/2} \frac{(\delta_0/2)^{\alpha_0/2}}{\Gamma(\alpha_0/2)} \frac{1}{|B_0|^{1/2}} \int \left\{ \left(\frac{1}{2\pi\sigma^2}\right)^{K/2} \right. \\
&\quad \left. \times \exp\left[-\frac{1}{2\sigma^2}(\beta-\bar\beta)'B_1^{-1}(\beta-\bar\beta)\right] d\beta \right\} \left\{ \left(\frac{1}{\sigma^2}\right)^{\alpha_1/2+1} \exp\left[-\frac{\delta_1}{2\sigma^2}\right] d\sigma^2 \right\},
\end{aligned}
$$

where we have used (4.5). Now integrate the first pair of curly braces as a multivariate normal and the second as an inverted gamma to find this expression for the marginal likelihood:

$$
\left(\frac{1}{\pi}\right)^{n/2} \frac{|B_1|^{1/2}}{|B_0|^{1/2}} \frac{\Gamma(\alpha_1/2)}{\Gamma(\alpha_0/2)} \frac{\delta_0^{\alpha_0/2}}{\delta_1^{\alpha_1/2}}.
$$

Context the item refers to (not part of the exercise; Greenberg §4.3, pp. 45–46, equations (4.2) and (4.6), reproduced verbatim).

$$
\begin{aligned}
\pi(\beta,\sigma^2) &= \pi(\beta|\sigma^2)\pi(\sigma^2) \\
&= N_K(\beta|\beta_0,\sigma^2 B_0)\, IG(\sigma^2|\alpha_0/2,\delta_0/2),
\end{aligned} \tag{4.2}
$$

[...] where the updated parameters are

$$
B_1 = (X'X + B_0^{-1})^{-1} \tag{4.6}
$$

$$\begin{aligned}\bar\beta &= B_1(X'y + B_0^{-1}\beta_0)\\ \alpha_1 &= \alpha_0 + n\\ \delta_1 &= \delta_0 + y'y + \beta_0'B_0^{-1}\beta_0 - \bar\beta' B_1^{-1}\bar\beta.\end{aligned}$$

Notation.

- y: $n\times 1$ response; X: $n\times K$ covariates of full column rank; $\beta\in\mathbb{R}^K$; $\sigma^2>0$.
- Prior hyperparameters: $\beta_0\in\mathbb{R}^K$, B_0 ($K\times K$, positive definite), $\alpha_0,\delta_0>0$; posterior ones $B_1,\bar\beta,\alpha_1,\delta_1$ from (4.6).
- $m(y)$: the marginal likelihood (the integral of the statement).

Step — first equality. The three factors of the integrand are the normal likelihood, the prior of β given σ^2 and the inverse-gamma prior; since $|\sigma^2 B_0|^{1/2} = (\sigma^2)^{K/2}|B_0|^{1/2}$,

$$f(y|\beta,\sigma^2) = (2\pi\sigma^2)^{-n/2}\exp\left[-\tfrac{1}{2\sigma^2}(y-X\beta)'(y-X\beta)\right], \tag{Q7.1}$$

$$\pi(\beta|\sigma^2) = (2\pi\sigma^2)^{-K/2}|B_0|^{-1/2}\exp\left[-\tfrac{1}{2\sigma^2}(\beta-\beta_0)'B_0^{-1}(\beta-\beta_0)\right], \tag{Q7.2}$$

$$\pi(\sigma^2) = \frac{(\delta_0/2)^{\alpha_0/2}}{\Gamma(\alpha_0/2)}(\sigma^2)^{-(\alpha_0/2+1)}\exp\left[-\tfrac{\delta_0}{2\sigma^2}\right]. \tag{Q7.3}$$

Their product is the first line of the statement.

Step — the completing-square identity behind (4.5). Expanding the three quadratic forms in the exponent,

$$\begin{aligned}&(y-X\beta)'(y-X\beta) + (\beta-\beta_0)'B_0^{-1}(\beta-\beta_0) + \delta_0\\ &\quad= \beta'(X'X + B_0^{-1})\beta - 2\beta'(X'y + B_0^{-1}\beta_0) + y'y + \beta_0'B_0^{-1}\beta_0 + \delta_0.\end{aligned} \tag{Q7.4}$$

By (4.6), $X'X + B_0^{-1} = B_1^{-1}$ and $X'y + B_0^{-1}\beta_0 = B_1^{-1}\bar\beta$; substituting into (Q7.4):

$$= \beta'(B_1^{-1})\beta - 2\beta'(B_1^{-1}\bar\beta) + y'y + \beta_0'B_0^{-1}\beta_0 + \delta_0. \tag{Q7.5}$$

Adding and subtracting $\bar\beta' B_1^{-1}\bar\beta$:

$$= (\beta-\bar\beta)'B_1^{-1}(\beta-\bar\beta) + [\delta_0 + y'y + \beta_0'B_0^{-1}\beta_0 - \bar\beta' B_1^{-1}\bar\beta] \tag{Q7.6}$$

$$= (\beta-\bar\beta)'B_1^{-1}(\beta-\bar\beta) + \delta_1, \tag{Q7.7}$$

using the definition of δ_1 in (4.6).

Step — second equality. Collect the constants of (Q7.1)–(Q7.3) outside, and the powers of σ^2 not in the $(2\pi\sigma^2)^{-K/2}$ factor:

$$(\sigma^2)^{-n/2}(\sigma^2)^{-(\alpha_0/2+1)} = (\sigma^2)^{-((\alpha_0+n)/2+1)} = (\sigma^2)^{-(\alpha_1/2+1)}, \tag{Q7.8}$$

using $\alpha_1 = \alpha_0 + n$. Substituting (Q7.7) for the exponent splits it into a β-part and a σ^2-part, which is the second line of the statement.

Step — integrate over β. The first brace is a $N_K(\bar\beta,\sigma^2 B_1)$ density without its factor $|B_1|^{-1/2}$; since that density integrates to one,

$$\int\left(\frac{1}{2\pi\sigma^2}\right)^{K/2}\exp\left[-\tfrac{1}{2\sigma^2}(\beta-\bar\beta)'B_1^{-1}(\beta-\bar\beta)\right]d\beta = |B_1|^{1/2}, \tag{Q7.9}$$

which no longer depends on σ^2.

Step — integrate over σ^2. The $IG(a,b)$ density integrates to one, so

$$\int_0^\infty (\sigma^2)^{-(a+1)}e^{-b/\sigma^2}d\sigma^2 = \frac{\Gamma(a)}{b^a}. \tag{Q7.10}$$

With $a = \alpha_1/2$, $b = \delta_1/2$ the second brace is $\Gamma(\alpha_1/2)/(\delta_1/2)^{\alpha_1/2}$.

Step — assemble. Substituting (Q7.9) and (Q7.10) into the second line of the statement:

$$m(y) = \left(\frac{1}{2\pi}\right)^{n/2} \frac{(\delta_0/2)^{\alpha_0/2}}{\Gamma(\alpha_0/2)\,|B_0|^{1/2}} \left(|B_1|^{1/2}\right) \left(\frac{\Gamma(\alpha_1/2)}{(\delta_1/2)^{\alpha_1/2}}\right). \tag{Q7.11}$$

The powers of 2 from $\delta_0/2$ and $\delta_1/2$ combine as

$$\frac{(1/2)^{\alpha_0/2}}{(1/2)^{\alpha_1/2}} = 2^{(\alpha_1-\alpha_0)/2} = 2^{n/2}, \tag{Q7.12}$$

$$\left(\frac{1}{2\pi}\right)^{n/2} 2^{n/2} = \left(\frac{1}{\pi}\right)^{n/2}. \tag{Q7.13}$$

Substituting (Q7.12)–(Q7.13) into (Q7.11):

$$m(y) = \left(\frac{1}{\pi}\right)^{n/2} \frac{|B_1|^{1/2}}{|B_0|^{1/2}} \frac{\Gamma(\alpha_1/2)}{\Gamma(\alpha_0/2)} \frac{\delta_0^{\alpha_0/2}}{\delta_1^{\alpha_1/2}}, \tag{Q7.14}$$

the expression of the statement. Independent check: under (4.2) the prior predictive of y is a multivariate $t_{\alpha_0}(X\beta_0, (\delta_0/\alpha_0)(I_n + XB_0X'))$; the script (function `g45`) evaluates both on simulated data and they agree to 10^{-9}.

Intuition. $|B_1|^{1/2}/|B_0|^{1/2}$ is an "Occam factor": the ratio of posterior to prior volume of β, small when the prior is diffuse relative to what the data pin down. δ_1 grows with the residual sum of squares plus prior–data conflict, so a poor fit lowers $m(y)$ through $\delta_1^{-\alpha_1/2}$.

Ref: Greenberg §4.3 (pp. 44–47) and §3.2.4 (marginal likelihood, Bayes factors); App. A.1.8, A.1.12.

Question 8. Greenberg 5.4 — accept–reject sampling of Beta(3, 3).

Source: Greenberg, E. (2007), *Introduction to Bayesian Econometrics*, Cambridge University Press, Exercise 5.4, p. 75 — reproduced verbatim.

> **5.4** Draw 500 observations from Beta(3, 3) using algorithm 5.3. Compute the mean and variance of the sample and compare them to the true values.
>
> *Context the item refers to (not part of the exercise; Greenberg §5.3, p. 67, Algorithm 5.3, reproduced verbatim).*
>
> **Algorithm 5.3: Beta distribution**
> **5.1** Draw u_1 and u_2 from $U(0, 1)$.
> **5.2** If
> $$u_2 \leq \frac{\text{Beta}(3,3)^{-1} u_1^2 (1-u_1)^2}{1.8750},$$
> return $y = u_1$ as a draw from the target distribution. Otherwise reject it and return to 1.

Notation.

- $f(y) = y^2(1-y)^2/B(3,3)$, $0 < y < 1$: Beta(3,3) target; $B(3,3) = \Gamma(3)\Gamma(3)/\Gamma(6) = 4/120 = 1/30$ (the book's "Beta$(3,3)^{-1}$").
- $g(y) = 1$ on $(0,1)$: proposal; c: bound with $f \leq cg$; $u_1, u_2 \sim U(0,1)$.

Step — the bound c. f is symmetric with its maximum at $y = 1/2$:

$$c = f(1/2) = 30\,(1/2)^2(1-1/2)^2 = \frac{30}{16} = 1.8750, \tag{Q8.1}$$

$$P(\text{accept}) = \frac{1}{c} = 0.5333. \tag{Q8.2}$$

Step — true moments. For Beta(a,b), $E = a/(a+b)$ and Var $= ab/[(a+b)^2(a+b+1)]$; with $a = b = 3$:

$$E(y) = \frac{3}{6} = 0.5, \tag{Q8.3}$$

$$\mathrm{Var}(y) = \frac{(3)(3)}{(6)^2(7)} = \frac{9}{252} = \frac{1}{28} = 0.03571. \tag{Q8.4}$$

Step — code (seed 54; function g54 of the check script).

```
rng = np.random.default_rng(54)
c = stats.beta.pdf(0.5, 3, 3)                # 1.8750
draws, tried = [], 0
while len(draws) < 500:
    u1, u2 = rng.uniform(), rng.uniform()
    tried += 1
    if u2 <= stats.beta.pdf(u1, 3, 3) / c:    # step 5.2: accept u1
        draws.append(u1)
d = np.array(draws); print(500 / tried, d.mean(), d.var(ddof=1))
```

Output: 900 proposals for 500 acceptances (rate $500/900 = 0.5556$ vs. 0.5333 in (Q8.2)); sample mean 0.5045, sample variance 0.03883.

Step — comparison. The mean error is $0.5045 - 0.5 = 0.0045$, about half of its standard error $\sqrt{(1/28)/500} = 0.0085$. The variance is $0.03883 - 0.03571 = 0.0031$ above (Q8.4), a sampling fluctuation of the same order (about 1.7 standard errors); both agree with the truth.

Intuition. The uniform proposal wastes about half the draws (those near 0 and 1 are accepted with low probability), but every accepted draw is an exact, independent Beta(3,3) draw: accept–reject trades computing time for exactness.

Ref: Greenberg §5.3 (pp. 66–69), Figure 5.1; Morettin & Singer §B.6.1 (pp. 468–470; Question 24 below).

Question 9. Greenberg 6.2 — the two-state Markov chain.
Source: Greenberg, E. (2007), *Introduction to Bayesian Econometrics*, Cambridge University Press, Exercise 6.2, pp. 87–88 — reproduced verbatim.

> **6.2** Let
>
> $$P = \begin{pmatrix} 1-\alpha & \alpha \\ \beta & 1-\beta \end{pmatrix}. \tag{6.10}$$
>
> **(a)** Find the invariant distribution if $0 < \alpha, \beta < 1$. Is the process aperiodic? Is it irreducible? What is $\lim P^n$ in this case?
>
> **(b)** Suppose $\alpha = \beta = 0$. Is there a unique invariant distribution? Is the process aperiodic? Is it irreducible? What is $\lim P^n$?
>
> **(c)** Suppose $\alpha = \beta = 1$. Is there a unique invariant distribution? Is the process aperiodic? Is it irreducible? What is $\lim P^n$?

Notation.

- States $\{1, 2\}$; $\alpha = P(X_{t+1} = 2 | X_t = 1)$, $\beta = P(X_{t+1} = 1 | X_t = 2)$, $\alpha, \beta \in [0, 1]$.
- $\pi = (\pi_1, \pi_2)$: row vector with $\pi_1 + \pi_2 = 1$, invariant if $\pi P = \pi$; $p_{ii}^{(n)}$: n-step return probability.

Step — (a) invariant distribution. The first column of $\pi P = \pi$ reads

$$\pi_1(1-\alpha) + \pi_2\beta = \pi_1, \tag{Q9.1}$$

$$\text{i.e.} \quad \pi_2\beta = \pi_1\alpha. \tag{Q9.2}$$

Substituting $\pi_2 = 1 - \pi_1$ into (Q9.2):

$$(1-\pi_1)\beta = \pi_1\alpha, \tag{Q9.3}$$

$$\pi_1 = \frac{\beta}{\alpha+\beta}, \qquad \pi_2 = \frac{\alpha}{\alpha+\beta}. \tag{Q9.4}$$

Step — (a) classification and limit. $p_{12} = \alpha > 0$ and $p_{21} = \beta > 0$, so $1 \leftrightarrow 2$: irreducible. $p_{11} = 1 - \alpha > 0$, so returns in one step are possible and the period is 1: aperiodic. The eigenvalues of P are 1 and $\lambda_2 = 1 - \alpha - \beta$ (trace $2 - \alpha - \beta$), and

$$P^n = \frac{1}{\alpha+\beta}\begin{pmatrix}\beta & \alpha\\ \beta & \alpha\end{pmatrix} + \frac{(1-\alpha-\beta)^n}{\alpha+\beta}\begin{pmatrix}\alpha & -\alpha\\ -\beta & \beta\end{pmatrix} \tag{Q9.5}$$

(at $n = 1$ the $(1,1)$ entry is $[\beta+\alpha(1-\alpha-\beta)]/(\alpha+\beta) = 1-\alpha$, as it should be). Since $|1-\alpha-\beta| < 1$,

$$\lim_{n\to\infty} P^n = \begin{pmatrix}\pi_1 & \pi_2\\ \pi_1 & \pi_2\end{pmatrix}. \tag{Q9.6}$$

Example: the matrix (6.2) is (Q9.5) with $\alpha = 0.25$, $\beta = 0.125$, so $\pi = (0.125/0.375,\ 0.25/0.375) = (0.333333,\ 0.666667)$, $\lambda_2 = 0.625$; the script finds both rows of P^{50} equal to $(0.333333, 0.666667)$.

Step — (b) $\alpha = \beta = 0$**.** $P = I$, so $\pi I = \pi$ for *every* distribution: the invariant distribution is not unique. Each state is absorbing and they do not communicate: reducible. $p_{ii} = 1$, so the period is 1: aperiodic. $P^n = I$ for all n, hence $\lim P^n = I$: the limit depends on the starting state.

Step — (c) $\alpha = \beta = 1$**.** $P = \begin{pmatrix}0 & 1\\ 1 & 0\end{pmatrix}$. (Q9.4) still applies ($\alpha + \beta = 2 > 0$): $\pi = (1/2, 1/2)$ is the unique invariant distribution. $1 \leftrightarrow 2$: irreducible. Returns occur only at even times ($p_{ii}^{(n)} > 0$ iff n is even): period 2, not aperiodic. With $1 - \alpha - \beta = -1$, (Q9.5) gives $P^n = I$ for even n and $P^n = P$ for odd n: $\lim P^n$ does not exist (only the Cesàro average $\frac{1}{N}\sum_{n\le N} P^n$ converges, to (Q9.6)).

Intuition. Uniqueness of π needs irreducibility; convergence of P^n to π needs aperiodicity as well. MCMC kernels are built to satisfy both, and $|\lambda_2|$ sets the speed: (6.2) forgets its start at rate 0.625^n.

Ref: Greenberg §6.1–6.2 (pp. 76–83); Tsay §12.1 (p. 614).

Question 10. Greenberg 7.1 — full conditionals of the normal model.
Source: Greenberg, E. (2007), *Introduction to Bayesian Econometrics*, Cambridge University Press, Exercise 7.1, p. 105 — reproduced verbatim.

> **7.1** Verify (7.1) and (7.2).
>
> *Context the item refers to (not part of the exercise; Greenberg §7.1.1, p. 93, reproduced verbatim).*
>
> Let $y_i \sim N(\mu, h^{-1})$, $i = 1, \ldots, n$, be independently distributed, where the distribution has been parameterized in terms of the precision. We assume the conditionally conjugate priors $\mu \sim N(\mu_0, h_0^{-1})$ and $h \sim G(\alpha_0/2, \delta_0/2)$. [...] From this specification, we have
>
> $$\pi(\mu, h|y) \propto h^{n/2} \exp\left[-\frac{h}{2}\sum(y_i - \mu)^2\right] \exp\left[-\frac{h_0}{2}(\mu - \mu_0)^2\right] \times h^{\alpha_0/2-1} \exp\left[-\frac{\delta_0 h}{2}\right].$$
>
> From here, it is easy to derive the conditional posterior distribution of h,
>
> $$\pi(h|\mu, y) \propto h^{(\alpha_0+n)/2-1} \exp\left[-h\frac{\delta_0 + \sum(y_i - \mu)^2}{2}\right]. \tag{7.1}$$
>
> [...] We complete the square in μ to obtain
>
> $$\pi(\mu|h, y) \propto \exp\left[-\frac{h_0 + hn}{2}\left(\mu - \frac{h_0\mu_0 + hn\bar{y}}{h_0 + hn}\right)^2\right], \tag{7.2}$$
>
> which should be recognized as $N[(h_0\mu_0 + hn\bar{y})/(h_0 + hn), (h_0 + hn)^{-1}]$ [...].

Notation.

- $y_i|\mu, h \sim N(\mu, 1/h)$; $h > 0$: precision; $\bar{y}$: sample mean.
- Priors: $\mu \sim N(\mu_0, 1/h_0)$, $h \sim G(\alpha_0/2, \delta_0/2)$, independent; $\mu_0 \in \mathbb{R}$, $h_0, \alpha_0, \delta_0 > 0$.

Step — joint posterior. The likelihood and the two prior kernels are

$$f(y|\mu,h)=\prod_{i=1}^{n}\Big(\frac{h}{2\pi}\Big)^{1/2}\exp\Big[-\frac{h}{2}(y_i-\mu)^2\Big]\propto h^{n/2}\exp\Big[-\frac{h}{2}\textstyle\sum_i(y_i-\mu)^2\Big], \tag{Q10.1}$$

$$\pi(\mu)\propto\exp\Big[-\frac{h_0}{2}(\mu-\mu_0)^2\Big],\qquad \pi(h)\propto h^{\alpha_0/2-1}\exp\Big[-\frac{\delta_0 h}{2}\Big], \tag{Q10.2}$$

whose product is the joint posterior displayed in the context.

Step — (7.1). Fixing μ, keep only the factors containing h:

$$\pi(h|\mu,y)\propto h^{n/2}\exp\Big[-\frac{h}{2}\textstyle\sum_i(y_i-\mu)^2\Big]\,h^{\alpha_0/2-1}\exp\Big[-\frac{\delta_0 h}{2}\Big] \tag{Q10.3}$$

$$=h^{(\alpha_0+n)/2-1}\exp\Big[-h\,\frac{\delta_0+\sum_i(y_i-\mu)^2}{2}\Big], \tag{Q10.4}$$

which is (7.1): the kernel of $G\big((\alpha_0+n)/2,\ (\delta_0+\sum_i(y_i-\mu)^2)/2\big)$.

Step — (7.2). Fixing h, keep the factors containing μ:

$$\pi(\mu|h,y)\propto\exp\Big[-\frac{h}{2}\textstyle\sum_i(y_i-\mu)^2-\frac{h_0}{2}(\mu-\mu_0)^2\Big]. \tag{Q10.5}$$

The decomposition $\sum_i(y_i-\mu)^2=\sum_i(y_i-\bar y)^2+n(\bar y-\mu)^2$ and dropping the first (free of μ) term give

$$\pi(\mu|h,y)\propto\exp\Big[-\tfrac{1}{2}\big\{hn(\mu-\bar y)^2+h_0(\mu-\mu_0)^2\big\}\Big]. \tag{Q10.6}$$

Expanding the braces of (Q10.6) and dropping constants:

$$hn(\mu^2-2\mu\bar y)+h_0(\mu^2-2\mu\mu_0)=(h_0+hn)\mu^2-2\mu(h_0\mu_0+hn\bar y) \tag{Q10.7}$$

$$=(h_0+hn)\Big(\mu-\frac{h_0\mu_0+hn\bar y}{h_0+hn}\Big)^2+\text{const.} \tag{Q10.8}$$

Substituting (Q10.8) into (Q10.6) gives (7.2), the kernel of $N\big[(h_0\mu_0+hn\bar y)/(h_0+hn),\ (h_0+hn)^{-1}\big]$.

Intuition. Given μ, each observation adds $1/2$ to the gamma shape and half a squared residual to its rate; given h, precisions add (h_0+hn) and the mean is the precision-weighted average of μ_0 and $\bar y$. These two standard draws are Algorithm 7.3.

Ref: Greenberg §7.1.1 (pp. 91–94); Tsay Results 12.1–12.2 (pp. 618–620).

Question 11. Greenberg 7.3 — Gibbs sampler for the coal-mining switch point.
Source: Greenberg, E. (2007), *Introduction to Bayesian Econometrics*, Cambridge University Press, Exercise 7.3, p. 106 — reproduced verbatim.

> **7.3** Construct a Gibbs algorithm to analyze the Poisson model with unknown switch point. Given the specification in Equations (7.3) and (7.4), show that
>
> $$\pi(\theta_1,\theta_2,k|y)\propto\theta_1^{\alpha_{10}-1}e^{-\beta_{10}\theta_1}\theta_2^{\alpha_{20}-1}e^{-\beta_{20}\theta_2}\prod_{i=1}^{k}e^{-\theta_1}\theta_1^{y_i}\prod_{i=k+1}^{n}e^{-\theta_2}\theta_2^{y_i},$$
>
> and verify that
>
> $$\theta_1|y,k\sim G\Big(\alpha_{10}+\sum_1^k y_i,\ \beta_{10}+k\Big),$$
>
> $$\theta_2|k,y\sim G\Big(\alpha_{20}+\sum_{k+1}^n y_i,\ \beta_{20}+n-k\Big),$$
>
> $$\pi(k|y,\theta_1,\theta_2)=\frac{e^{k(\theta_2-\theta_1)}(\theta_1/\theta_2)^{\sum_1^k y_i}}{\sum_{k=1}^n e^{k(\theta_2-\theta_1)}(\theta_1/\theta_2)^{\sum_1^k y_i}},\qquad k=1,\dots,n.$$

Apply your algorithm to the mining disaster data analyzed by Carlin et al. (1992a); the data may be found in the article. Your analysis should generate marginal posterior distributions for θ_1, θ_2, $p(k)$, and the marginal likelihood computed by the method of Chib (1995). For simplicity, set $\alpha_{10} = \alpha_{20} = 0.5$ and $\beta_{10} = \beta_{20} = 1.0$. (Curiously, although these data have been analyzed from the Bayesian viewpoint in several papers, none of them present historical material that might shed light on when changes in parameters took place. For example, changes in government regulation, technology, or the location of mines might be expected to affect the probability of a disaster. A more general version of the model permits multiple switch points if necessary.)

Context the item refers to (not part of the exercise; Greenberg §7.1.1, pp. 93–94, equations (7.3)–(7.4), reproduced verbatim).

$$p(y_i) = \begin{cases} \dfrac{e^{-\theta_1}\theta_1^{y_i}}{y_i!}, & \text{for } i = 1, \ldots, k, \\ \dfrac{e^{-\theta_2}\theta_2^{y_i}}{y_i!}, & \text{for } i = k+1, \ldots, n, \end{cases} \tag{7.3}$$

where $y_i = 0, 1, \ldots$, and the switch point k is unknown. The specification is completed by assigning the conditionally conjugate priors,

$$\theta_1 \sim G(\alpha_{10}, \beta_{10}), \quad \theta_2 \sim G(\alpha_{20}, \beta_{20}), \quad \pi(k = j) = 1/n, \ j = 1, \ldots, n. \tag{7.4}$$

Notation.

- y_i: number of British coal-mining disasters in year $1850+i$, $i = 1, \ldots, n$, $n = 112$ (1851–1962), $\sum_i y_i = 191$; data of Carlin, Gelfand and Smith (1992a), embedded in the check script (its first 40 years total 125, as on the lecturer's slides).
- $k \in \{1, \ldots, n\}$: last year of regime 1 (year $1850 + k$); $\theta_1, \theta_2 > 0$: disaster rates before/after.
- $S_k = \sum_{i=1}^{k} y_i$ ($S_0 = 0$, $S_n = 191$); $\alpha_{j0} = 0.5$, $\beta_{j0} = 1$; G: number of retained draws.

Step — joint posterior. By Bayes theorem with (7.3)–(7.4),

$$\pi(\theta_1, \theta_2, k|y) \propto \prod_{i=1}^{k} \frac{e^{-\theta_1}\theta_1^{y_i}}{y_i!} \prod_{i=k+1}^{n} \frac{e^{-\theta_2}\theta_2^{y_i}}{y_i!}\ \theta_1^{\alpha_{10}-1} e^{-\beta_{10}\theta_1}\ \theta_2^{\alpha_{20}-1} e^{-\beta_{20}\theta_2}\ \frac{1}{n}. \tag{Q11.1}$$

Dropping $\prod_i y_i!$ and $1/n$ (free of the parameters) gives the expression of the statement.

Step — $\theta_1|y, k$ and $\theta_2|y, k$. Keeping in (Q11.1) only the factors with θ_1:

$$\pi(\theta_1|y, k, \theta_2) \propto \theta_1^{\alpha_{10}-1} e^{-\beta_{10}\theta_1} \prod_{i=1}^{k} e^{-\theta_1}\theta_1^{y_i} \tag{Q11.2}$$

$$= \theta_1^{\alpha_{10}+S_k-1} e^{-(\beta_{10}+k)\theta_1}, \tag{Q11.3}$$

the kernel of $G(\alpha_{10} + S_k, \beta_{10} + k)$, free of θ_2. In the same way,

$$\pi(\theta_2|y, k, \theta_1) \propto \theta_2^{\alpha_{20}+S_n-S_k-1} e^{-(\beta_{20}+n-k)\theta_2}. \tag{Q11.4}$$

Step — $k|y, \theta_1, \theta_2$. Keeping the factors of (Q11.1) that depend on k:

$$\pi(k|y, \theta_1, \theta_2) \propto e^{-k\theta_1}\theta_1^{S_k}\, e^{-(n-k)\theta_2}\theta_2^{S_n-S_k}. \tag{Q11.5}$$

Dividing (Q11.5) by $e^{-n\theta_2}\theta_2^{S_n}$, which does not depend on k:

$$\pi(k|y, \theta_1, \theta_2) \propto e^{k(\theta_2-\theta_1)}(\theta_1/\theta_2)^{S_k}, \qquad k = 1, \ldots, n, \tag{Q11.6}$$

and normalising over k gives the discrete distribution of the statement.

Step — Gibbs algorithm. Start at $k^{(0)} = 56$; at iteration g draw $\theta_1^{(g)}$ from (Q11.3) with $k = k^{(g-1)}$, $\theta_2^{(g)}$ from (Q11.4), and $k^{(g)}$ from (Q11.6) (a categorical draw over $1, \ldots, n$). Burn-in 1,000, keep $G = 10{,}000$, seed 73 (function `g73`):

```
S = np.concatenate([[0], np.cumsum(y)]); k_grid = np.arange(1, n + 1); k = n // 2
for g in range(G + burn):
    t1 = rng.gamma(a10 + S[k], 1 / (b10 + k))              # (Q11.3)
    t2 = rng.gamma(a20 + S[n] - S[k], 1 / (b20 + n - k))   # (Q11.4)
    logp = k_grid * (t2 - t1) + S[1:] * np.log(t1 / t2)    # (Q11.6)
    p = np.exp(logp - logp.max()); p /= p.sum()
    k = rng.choice(k_grid, p=p)
```

Step — exact benchmark. Integrating θ_1, θ_2 out of (Q11.1) with the gamma integral (Q5.3) (Question 5):

$$p(y|k) = \frac{\beta_{10}^{\alpha_{10}}}{\Gamma(\alpha_{10})} \frac{\Gamma(\alpha_{10} + S_k)}{(\beta_{10} + k)^{\alpha_{10}+S_k}} \cdot \frac{\beta_{20}^{\alpha_{20}}}{\Gamma(\alpha_{20})} \frac{\Gamma(\alpha_{20} + S_n - S_k)}{(\beta_{20} + n - k)^{\alpha_{20}+S_n-S_k}} \cdot \frac{1}{\prod_i y_i!}, \tag{Q11.7}$$

$$p(k|y) = \frac{p(y|k)}{\sum_{j=1}^{n} p(y|j)}, \qquad m(y) = \frac{1}{n}\sum_{k=1}^{n} p(y|k). \tag{Q11.8}$$

Step — Chib (1995) marginal likelihood. For any point $\theta^* = (\theta_1^*, \theta_2^*)$, Bayes theorem rearranged gives the identity

$$\log m(y) = \log f(y|\theta^*) + \log \pi(\theta^*) - \log \pi(\theta^*|y), \tag{Q11.9}$$

$$f(y|\theta_1, \theta_2) = \frac{1}{n}\sum_{k=1}^{n} f(y|\theta_1, \theta_2, k), \tag{Q11.10}$$

$$\hat{\pi}(\theta^*|y) = \frac{1}{G}\sum_{g=1}^{G} G\big(\theta_1^*|\alpha_{10} + S_{k^{(g)}}, \beta_{10} + k^{(g)}\big)\, G\big(\theta_2^*|\alpha_{20} + S_n - S_{k^{(g)}}, \beta_{20} + n - k^{(g)}\big), \tag{Q11.11}$$

where (Q11.11) uses that, given k, θ_1 and θ_2 are independent with the gamma conditionals (Q11.3)–(Q11.4). Take θ^* = posterior means.

Step — results. Posterior summaries ($G = 10{,}000$; exact values from (Q11.8)):

	mean (MCMC)	mean (exact)	sd	95% interval
θ_1 (1851–1850 + k)	3.0512	3.0500	0.2861	(2.513, 3.628)
θ_2 (1851 + k–1962)	0.9144	0.9147	0.1170	(0.702, 1.157)

k (last year of regime 1)	41 (1891)	40 (1890)	39 (1889)	42 (1892)
$p(k\|y)$, MCMC	0.2514	0.1903	0.1374	0.0991
$p(k\|y)$, exact	0.2482	0.1848	0.1414	0.1010

Chib: $\log f(y|\theta^*) = -172.2568$, $\log \pi(\theta^*) = -5.6234$, $\log \hat{\pi}(\theta^*|y) = 1.5790$. Substituting into (Q11.9):

$$\log \hat{m}(y) = (-172.2568) + (-5.6234) - (1.5790) \tag{Q11.12}$$

$$= -179.4592, \tag{Q11.13}$$

against the exact $\log m(y) = -179.4590$ from (Q11.8).

Intuition. The disaster rate falls from about 3.05 to about 0.91 per year, with the switch most likely right after 1889–1892 (posterior mode: 1891). Because the gamma conditionals are known in closed form, the Rao–Blackwellised ordinate (Q11.11) is very accurate, and Chib's estimate lands within 0.0002 of the exact log marginal likelihood.

Ref: Greenberg §7.1.1 (pp. 93–94), §7.1.2 (marginal likelihood by Chib's method, p. 95); Bolfarine & Sandoval §3.4 (profile likelihood, used for the same data in class 1).

Question 12. Greenberg 7.5 — random-walk Metropolis–Hastings for the Laplace distribution.
Source: Greenberg, E. (2007), *Introduction to Bayesian Econometrics*, Cambridge University Press, Exercise 7.5, p. 106 — reproduced verbatim.

> **7.5** Construct a random walk MH sampler to generate a sample of 10,000 from the Laplace distribution, $f(x) = (1/2)\, e^{-|x|}$, $-\infty < x < \infty$. Use $u \sim N(0, \sigma^2)$ to generate proposals $y = x^{(g-1)} + u$. Compare the mean, variances, and autocorrelations of the generated series for various values of σ^2.

Notation.

- $f(x) = \frac{1}{2}e^{-|x|}$: target, mean 0, variance 2; $x^{(g)}$: state at iteration g; y: proposal; $u \sim N(0, \sigma^2)$.
- $\alpha(x, y)$: acceptance probability; r_L: lag-L sample autocorrelation of the retained chain.

Step — acceptance probability. The random-walk proposal density $q(x, y) = \phi((y - x)/\sigma)/\sigma$ is symmetric, $q(x, y) = q(y, x)$, so it cancels from the MH ratio:

$$\alpha(x, y) = \min\left\{1, \frac{f(y)q(y, x)}{f(x)q(x, y)}\right\} = \min\left\{1, \frac{f(y)}{f(x)}\right\} \tag{Q12.1}$$

$$= \min\left\{1, \exp\left(|x| - |y|\right)\right\}. \tag{Q12.2}$$

Step — true moments. $E(x) = 0$ by symmetry and $\mathrm{Var}(x) = \int x^2 \frac{1}{2} e^{-|x|} dx = \int_0^\infty x^2 e^{-x} dx = \Gamma(3) = 2$.

Step — code (seed 75; function `g75`). Start at $x = 0$, burn-in 1,000, keep 10,000 per σ^2:

```
for s2 in (0.1, 1.0, 10.0, 100.0):
    x, acc, out = 0.0, 0, np.empty(G)
    for g in range(G + burn):
        prop = x + rng.normal(0, s2 ** 0.5)
        if np.log(rng.uniform()) <= abs(x) - abs(prop):   # (Q12.2)
            x = prop; acc += g >= burn
        if g >= burn: out[g - burn] = x
```

Step — results.

σ^2	acceptance	mean	variance	r_1	r_{10}
0.1	0.8835	−0.306	2.317	0.983	0.852
1	0.6970	−0.097	1.908	0.880	0.285
10	0.3928	−0.056	1.976	0.668	0.025
100	0.1577	−0.062	2.120	0.714	0.018
truth	—	0	2	0	0

A tiny step ($\sigma^2 = 0.1$) is almost always accepted but barely moves: $r_1 = 0.983$, and after 10,000 draws the mean is still −0.306. A huge step ($\sigma^2 = 100$) is usually rejected (acceptance 0.158), so the chain repeats values and r_1 climbs back to 0.714. $\sigma^2 = 10$ (acceptance 0.39) has the smallest lag-1 autocorrelation and mean and variance closest to the truth.

Intuition. Every σ^2 gives a chain with the right invariant distribution; what σ^2 controls is efficiency. The best random-walk scale balances step length against acceptance, which is why tuning to an acceptance rate of roughly 0.3–0.5 in one dimension is the standard advice.

Ref: Greenberg §7.2.1 (pp. 96–101) and §7.3 (autocorrelation, n.s.e., p. 102); Tsay §12.4.1–12.4.2 (pp. 622–623).

Question 13. Greenberg 7.8 — MH within Gibbs with a tailored truncated-t proposal.
Source: Greenberg, E. (2007), *Introduction to Bayesian Econometrics*, Cambridge University Press, Exercise 7.8, p. 107 — reproduced verbatim.

> **7.8** Generalize Exercise 7.7 to allow for an unknown variance, $u_i \sim N(0, \sigma^2)$, with prior distribution $\sigma^2 \sim \text{IG}(5/2, 3/2)$. Verify that $\sigma^2|y, \beta$ has an inverse gamma distribution, but that the distribution of $\beta|y, \sigma^2$ is not standard. Devise a two-block algorithm in which $\sigma^2|y, \beta$ is generated in a Gibbs step and β is generated by a tailored kernel where $\hat{\beta}$ depends on σ^2, so that the mean and scale factor of the truncated Student-t distribution must be recomputed at each step. Choose a value of σ^2 from its prior distribution, and generate and analyze a data set as in Exercise 7.7.
>
> *Context the item refers to (not part of the exercise; Greenberg, Exercise 7.7, pp. 106–107, reproduced verbatim).*
>
> **7.7** Consider the model
>
> $$y_i = \beta x_i + u_i, \quad u_i \sim N(0,1), \quad i = 1, \ldots, n,$$
>
> with the gamma prior distribution $\beta \sim G(2,1)$, $\beta > 0$. Verify the posterior distribution
>
> $$\pi(\beta|y) \propto \beta \exp[-\beta] \exp\left[-\frac{1}{2}\sum_{i=1}^{n}(y_i - \beta x_i)^2\right] 1(\beta > 0).$$
>
> Note that this distribution does not have a standard form. Construct an MH algorithm to sample from this distribution with an independence kernel, where the kernel is a Student-t distribution truncated to the region $(0, \infty)$, with five degrees of freedom, mean equal to the value of β that maximizes the posterior distribution ($\hat{\beta}$), and scale factor equal to the negative inverse of the second derivative of the log posterior distribution evaluated at $\hat{\beta}$. Verify that
>
> $$\hat{\beta} = \frac{\left(\sum x_i y_i - 1\right) + \sqrt{\left(\sum x_i y_i - 1\right)^2 + 4\sum x_i^2}}{2\sum x_i^2}$$
>
> and that the scale factor is $(1/\hat{\beta}^2 + \sum x_i^2)^{-1}$. Generate a data set by choosing $n = 50$, x_i from $N(0,1)$, and a value of β from its prior distribution. Write a program to implement your algorithm and see how well β is determined. You may try larger values of n to explore the effect of sample size, and, depending on the acceptance rate, you may wish to change the scale factor.

Notation.

- $y_i = \beta x_i + u_i$, $u_i \sim N(0, \sigma^2)$, $i = 1, \ldots, n$ ($n = 50$); $S_{xx} = \sum x_i^2$, $S_{xy} = \sum x_i y_i$, $S_{yy} = \sum y_i^2$; $\text{SSR}(\beta) = \sum_i (y_i - \beta x_i)^2$.
- Priors: $\beta \sim G(2,1)$, $\pi(\beta) = \beta e^{-\beta} 1(\beta > 0)$; $\sigma^2 \sim IG(5/2, 3/2)$, $\pi(\sigma^2) \propto (\sigma^2)^{-7/2} e^{-3/(2\sigma^2)}$; independent.
- $\hat{\beta}(\sigma^2)$, $V(\sigma^2)$: conditional posterior mode and scale; $t_5(m, V)$: Student-t, 5 d.f., location m, scale^2 V.

Step — joint posterior. Likelihood times the two priors:

$$\pi(\beta, \sigma^2|y) \propto (\sigma^2)^{-n/2} \exp\Big[-\frac{\text{SSR}(\beta)}{2\sigma^2}\Big] \beta e^{-\beta} 1(\beta > 0) \, (\sigma^2)^{-(5/2+1)} e^{-(3/2)/\sigma^2}. \tag{Q13.1}$$

Step — $\sigma^2|y, \beta$ is inverse gamma. Keeping the factors of (Q13.1) with σ^2:

$$\pi(\sigma^2|y, \beta) \propto (\sigma^2)^{-(5/2+n/2+1)} \exp\Big[-\frac{3/2 + \text{SSR}(\beta)/2}{\sigma^2}\Big], \tag{Q13.2}$$

$$\sigma^2|y, \beta \sim IG\Big(\frac{5}{2} + \frac{n}{2}, \; \frac{3}{2} + \frac{\text{SSR}(\beta)}{2}\Big). \tag{Q13.3}$$

Step — $\beta|y, \sigma^2$ is not standard. Keeping the factors with β and expanding $\text{SSR}(\beta) = S_{yy} - 2\beta S_{xy} + \beta^2 S_{xx}$:

$$\log \pi(\beta|y, \sigma^2) = \log \beta - \beta - \frac{S_{yy} - 2\beta S_{xy} + \beta^2 S_{xx}}{2\sigma^2} + \text{const}, \quad \beta > 0. \tag{Q13.4}$$

The $\log\beta$ term (from the gamma prior) next to a quadratic (from the normal likelihood) is neither a normal nor a gamma log-kernel: no standard distribution, hence an MH step. (With $\sigma^2 = 1$, exponentiating (Q13.4) is the posterior of Exercise 7.7.)

Step — mode $\hat\beta(\sigma^2)$. Differentiating (Q13.4):

$$\frac{d\log\pi}{d\beta} = \frac{1}{\beta} - 1 + \frac{S_{xy} - \beta S_{xx}}{\sigma^2} = 0. \tag{Q13.5}$$

Multiplying (Q13.5) by $\beta\sigma^2 > 0$:

$$\sigma^2 - \sigma^2\beta + \beta S_{xy} - \beta^2 S_{xx} = 0, \tag{Q13.6}$$

$$S_{xx}\beta^2 - (S_{xy} - \sigma^2)\beta - \sigma^2 = 0. \tag{Q13.7}$$

The product of the roots of (Q13.7) is $-\sigma^2/S_{xx} < 0$, so exactly one is positive:

$$\hat\beta(\sigma^2) = \frac{(S_{xy} - \sigma^2) + \sqrt{(S_{xy} - \sigma^2)^2 + 4\sigma^2 S_{xx}}}{2S_{xx}}, \tag{Q13.8}$$

which with $\sigma^2 = 1$ is the formula of Exercise 7.7.

Step — scale $V(\sigma^2)$. Differentiating (Q13.5) again and taking minus the inverse at $\hat\beta$:

$$\frac{d^2\log\pi}{d\beta^2} = -\frac{1}{\beta^2} - \frac{S_{xx}}{\sigma^2}, \tag{Q13.9}$$

$$V(\sigma^2) = \Big(\frac{1}{\hat\beta^2} + \frac{S_{xx}}{\sigma^2}\Big)^{-1}, \tag{Q13.10}$$

equal to $(1/\hat\beta^2 + \sum x_i^2)^{-1}$ when $\sigma^2 = 1$, as Exercise 7.7 states. The script checks (Q13.8)–(Q13.10) by finite differences.

Step — two-block algorithm. Given $(\beta^{(g-1)}, \sigma^{2(g-1)})$:

1. Gibbs step: draw $\sigma^{2(g)}$ from (Q13.3) with $\beta = \beta^{(g-1)}$.
2. Recompute $\hat\beta$ and V from (Q13.8)–(Q13.10) at $\sigma^{2(g)}$; draw $\beta' \sim t_5(\hat\beta, V)$ truncated to $(0,\infty)$ (inverse c.d.f.: $u \sim U(F_5(-\hat\beta/\sqrt{V}), 1)$, $\beta' = \hat\beta + \sqrt{V}F_5^{-1}(u)$).
3. Accept $\beta^{(g)} = \beta'$ with probability

$$\alpha = \min\left\{1, \frac{\pi(\beta'|y,\sigma^{2(g)})\, q(\beta^{(g-1)})}{\pi(\beta^{(g-1)}|y,\sigma^{2(g)})\, q(\beta')}\right\}, \qquad q(b) \propto t_5 \text{ density at } (b-\hat\beta)/\sqrt{V}, \tag{Q13.11}$$

else $\beta^{(g)} = \beta^{(g-1)}$. The truncation constant is common to both q terms and cancels.

The proposal may depend on σ^2 because, inside the β block, σ^2 is held fixed: step 2–3 is an independence MH kernel whose invariant distribution is $\pi(\beta|y,\sigma^2)$.

Step — code (seed 78; function `g78`).

```
beta_true = rng.gamma(2.0, 1.0); s2_true = 1 / rng.gamma(2.5, 1 / 1.5)
x = rng.normal(size=n); y = beta_true * x + rng.normal(0, s2_true ** 0.5, n)
for g in range(G + burn):
    s2 = 1 / rng.gamma(2.5 + n / 2, 1 / (1.5 + (Syy - 2*b*Sxy + b*b*Sxx) / 2))  # (Q13.3)
    bh, V = mode_scale(s2); sd = V ** 0.5                                       # (Q13.8),(Q13.10)
    prop = bh + sd * stats.t.ppf(rng.uniform(stats.t.cdf(-bh / sd, 5), 1.0), 5)
    logq = lambda v: stats.t.logpdf((v - bh) / sd, 5)
    if np.log(rng.uniform()) <= logpost(prop, s2) - logpost(b, s2) + logq(b) - logq(prop):
        b = prop                                                                # (Q13.11)
```

Step — results. Data: $\beta = 0.9369$ drawn from $G(2,1)$, $\sigma^2 = 0.9079$ drawn from $IG(5/2, 3/2)$, $n = 50$; OLS $\hat\beta = 0.9788$, $s^2 = 1.0762$. With burn-in 1,000 and $G = 10{,}000$: $E(\beta|y) = 0.9806$ (sd 0.1785, 95% interval $(0.6329, 1.3353)$); $E(\sigma^2|y) = 1.0702$ (sd 0.2121, interval $(0.7373, 1.5616)$);

acceptance rate of the β step 0.9267. Both true values lie inside their intervals; with $n = 50$ the data dominate both priors, so the posterior means sit next to OLS.

Intuition. A tailored proposal (mode and curvature of the target) makes the independence kernel almost the target itself, hence the 93% acceptance. Re-tailoring at every σ^2 costs one quadratic formula per iteration and keeps that efficiency as σ^2 moves.

Ref: Greenberg §7.2.1 (tailored kernels, pp. 99–100; Algorithm 7.6, MH with two blocks, p. 100); Tsay §12.4.2.

Question 14. Greenberg 8.1 — Bayesian regression for birth weight.
Source: Greenberg, E. (2007), *Introduction to Bayesian Econometrics*, Cambridge University Press, Exercise 8.1, p. 132 — reproduced verbatim.

> **8.1** Download the data in "Birth weight II" from the StatLabs Web site, www.stat.berkeley.edu/users/statlabs/labs.html. The variables are birth weight (ounces, 999 unknown), gestation (length of pregnancy in days, 999 unknown), parity (0 = firstborn, 9 is unknown), age (mother's age in years), height (mother's height in inches), weight (mother's prepregnancy weight in pounds), smoke (0 not now, 1 yes now, 9 = unknown).
>
> Specify a linear regression, where the response variable is birth weight, and the covariates are a constant and the remaining variables in the data set. Eliminate observations for which you do not have complete information on the variables. Specify prior distributions for the coefficients and the variance (or precision). Summarize the results for the dummy variable for smoking. Does smoking by the mother appear to affect the baby's birth weight?

Notation.

- y_i: birth weight (oz); $x_i = (1, \text{gestation}, \text{parity}, \text{age}, \text{height}, \text{weight}, \text{smoke})'$, $K = 7$; X: $n \times K$.
- $\beta \in \mathbb{R}^7$, $\sigma^2 > 0$; priors $\beta \sim N_7(\beta_0, B_0)$, $\sigma^2 \sim IG(\alpha_0/2, \delta_0/2)$ as in (8.1); β_{smoke}: 7th coefficient.

Step — data. File `babies.data` (StatLabs, saved to `Listas/data/`, provenance in its README): 1,236 births. Missing codes in the file: gestation 999, age 99, height 99, weight 999, smoke 9 (birth weight has none; parity is coded 0/1 with no 9s). Dropping the 62 incomplete rows leaves $n = 1{,}174$.

Step — priors. $\beta_0 = 0$, $B_0 = 10^4 I_7$ (prior s.d. 100 for every coefficient, very diffuse at the scale of these covariates); $\alpha_0 = 6$, $\delta_0 = 1800$, so $E(\sigma^2) = \delta_0/(\alpha_0 - 2) = 450$ (birth-weight s.d. about 21 oz), worth 6 pseudo-observations against 1,174.

Step — Gibbs sampler. Greenberg (8.2)–(8.3):

$$\beta|\sigma^2, y \sim N_K(\bar\beta, B_1), \quad B_1 = (\sigma^{-2}X'X + B_0^{-1})^{-1}, \quad \bar\beta = B_1(\sigma^{-2}X'y + B_0^{-1}\beta_0), \tag{Q14.1}$$

$$\sigma^2|\beta, y \sim IG\big(\alpha_1/2, \delta_1/2\big), \quad \alpha_1 = \alpha_0 + n, \quad \delta_1 = \delta_0 + (y - X\beta)'(y - X\beta). \tag{Q14.2}$$

Burn-in 1,000, $G = 10{,}000$, seed 81 (function `g81`):

```
for g in range(G + burn):
    B1 = np.linalg.inv(XtX / s2 + B0inv); bbar = B1 @ (Xty / s2 + B0inv @ b0)
    beta = bbar + np.linalg.cholesky(B1) @ rng.normal(size=K)         # (Q14.1)
    r = y - X @ beta
    s2 = 1 / rng.gamma((a0 + n) / 2, 2 / (d0 + r @ r))                # (Q14.2)
```

Step — results.

	post. mean	post. sd	95% interval		OLS (s.e.)
constant	−78.6033	14.1929	−106.1171	−51.1482	−80.4109 (14.3466)
gestation (days)	0.4421	0.0292	0.3845	0.4982	0.4440 (0.0291)
parity	−3.3401	1.1372	−5.5466	−1.1178	−3.3272 (1.1289)
age (years)	−0.0114	0.0864	−0.1791	0.1587	−0.0090 (0.0858)
height (in)	1.1342	0.2039	0.7389	1.5368	1.1540 (0.2050)
weight (lb)	0.0508	0.0254	0.0008	0.1019	0.0502 (0.0252)
smoke	**−8.4084**	**0.9465**	**−10.2359**	**−6.5505**	−8.4007 (0.9538)
σ^2	251.1114	10.3514	231.6747	272.2228	

For the smoking dummy, $P(\beta_{\text{smoke}} < 0|y) = 1.0000$ (all 10,000 draws are negative). The only visible prior effect is on the constant (-78.6 vs. -80.4), shrunk slightly toward 0 by the s.d.-100 prior; all slopes match OLS.

Step — answer. Yes. Holding gestation, parity, age, height and pre-pregnancy weight fixed, babies of mothers who smoke weigh on average 8.41 oz less, i.e. $(-8.4084)(28.3495) = -238.37$ g, with 95% posterior probability between 6.55 and 10.24 oz less.

Intuition. With 1,174 observations and a diffuse prior the Bayesian and OLS answers coincide; the Bayesian output adds a direct probability statement ($P(\beta_{\text{smoke}} < 0|y) \approx 1$). Two cautions: the comparison is observational (smokers may differ in unmeasured ways; classes 5+6), and because smoking also shortens gestation, controlling for gestation leaves only the part of the smoking effect that does not run through it.

Ref: Greenberg §8.1.1 (pp. 111–114), §4.9; Wooldridge §3.2 (`BWGHT` is a related data set).

Question 15. Greenberg 8.8 — logit posterior and latent data.
Source: Greenberg, E. (2007), *Introduction to Bayesian Econometrics*, Cambridge University Press, Exercise 8.8, pp. 132–133 — reproduced verbatim.

> **8.8** Verify that the posterior distribution of the logit model with a normal prior (Section 8.2.3) has the form
>
> $$\pi(\beta|y) \propto \prod \left(\frac{\exp[x_i'\beta]}{1+\exp[x_i'\beta]}\right)^{y_i} \left(\frac{1}{1+\exp[x_i'\beta]}\right)^{1-y_i} \times \exp\left[-\frac{1}{2}(\beta-\beta_0)'B_0^{-1}(\beta-\beta_0)\right],$$
>
> and show that including the latent data in the sampler does not simplify the posterior distribution.
>
> *Context the item refers to (not part of the exercise; Greenberg §8.2.3, pp. 126–127, reproduced verbatim).*
>
> The binary logit model is another model for binary data. It is based on
>
> $$P(y_i = 1) = P_i = \frac{\exp[x_i'\beta]}{1+\exp[x_i'\beta]},$$
>
> [...] This model can be written in the form of latent data as
>
> $$y_i^* = x_i'\beta + u_i,$$
>
> where u_i has the *logistic* distribution, $P(U_i \le u_i) = \Lambda(u_i) = e^{u_i}/(1+e^{u_i})$ and density function $\lambda(u_i) = e^{u_i}/(1+e^{u_i})^2$.

Notation.

- $y_i \in \{0,1\}$, $x_i \in \mathbb{R}^K$, $i = 1,\dots,n$; $\beta \in \mathbb{R}^K$ with prior $N_K(\beta_0, B_0)$.
- Λ, λ: logistic c.d.f. and density; y_i^*: latent utility, $y_i = 1(y_i^* > 0)$.

Step — posterior. Each y_i is Bernoulli(P_i), and

$$1 - P_i = 1 - \frac{\exp[x_i'\beta]}{1+\exp[x_i'\beta]} = \frac{1}{1+\exp[x_i'\beta]}, \tag{Q15.1}$$

$$f(y|\beta) = \prod_{i=1}^{n} P_i^{y_i}(1-P_i)^{1-y_i}, \tag{Q15.2}$$

$$\pi(\beta) \propto \exp\left[-\tfrac{1}{2}(\beta-\beta_0)'B_0^{-1}(\beta-\beta_0)\right]. \tag{Q15.3}$$

Substituting P_i and (Q15.1) into (Q15.2) and multiplying by (Q15.3) gives exactly the displayed $\pi(\beta|y)$. It is not a normal or any standard density in β (the log-likelihood is not quadratic in β), so Greenberg samples it by MH.

Step — latent-data representation. With $y_i^* = x_i'\beta + u_i$ and $y_i = 1(y_i^* > 0)$, by symmetry of the logistic distribution,

$$P(y_i = 1|\beta) = P(u_i > -x_i'\beta) = 1 - \Lambda(-x_i'\beta) = \Lambda(x_i'\beta) = P_i, \tag{Q15.4}$$

so the representation reproduces the model.

Step — augmented posterior. Treating y^* as parameters,

$$\pi(\beta, y^*|y) \propto \pi(\beta) \prod_{i=1}^{n} \lambda(y_i^* - x_i'\beta)\big[y_i 1(y_i^* > 0) + (1 - y_i)1(y_i^* \le 0)\big]. \tag{Q15.5}$$

The conditional of y^* is easy (independent truncated logistics). The conditional of β keeps the factors of (Q15.5) with β:

$$\pi(\beta|y^*, y) \propto \exp\Big[-\tfrac{1}{2}(\beta - \beta_0)' B_0^{-1}(\beta - \beta_0)\Big] \prod_{i=1}^{n} \frac{e^{y_i^* - x_i'\beta}}{\big(1 + e^{y_i^* - x_i'\beta}\big)^2}. \tag{Q15.6}$$

The log of each logistic factor, $(y_i^* - x_i'\beta) - 2\log(1 + e^{y_i^* - x_i'\beta})$, is not quadratic in β, so (Q15.6) is no more standard than $\pi(\beta|y)$: an MH step is still needed, now in a space enlarged by n latent variables. Integrating y_i^* out of (Q15.5) returns the original form, since for $y_i = 1$

$$\int_0^\infty \lambda(y_i^* - x_i'\beta)\, dy_i^* = 1 - \Lambda(-x_i'\beta) = \Lambda(x_i'\beta) = P_i, \tag{Q15.7}$$

and $\int_{-\infty}^0 \lambda(y_i^* - x_i'\beta) dy_i^* = 1 - P_i$ for $y_i = 0$.

Intuition. Latent data help when the latent model is conditionally conjugate. In the probit, $u_i \sim N(0,1)$ makes $\beta|y^*$ a normal regression posterior (Gibbs, §8.2.2); with logistic errors the latent model is itself a non-Gaussian regression, so augmentation adds work without removing the MH step.

Ref: Greenberg §8.2.2–8.2.3 (pp. 122–129).

Block 2 — Tsay (2010) *(Ch. 12 • 3 questions)*

Question 16. Tsay 12.1 — normal mean, one observation.

Source: Tsay, R. S. (2010), *Analysis of Financial Time Series*, 3rd ed., Wiley, Exercise 12.1, p. 670 — reproduced verbatim.

> **12.1.** Suppose that x is normally distributed with mean μ and variance 4. Assume that the prior distribution of μ is also normal with mean 0 and variance 25. What is the posterior distribution of μ given the data point x?

Notation.

- $x|\mu \sim N(\mu, \sigma^2)$, $\sigma^2 = 4$, one observation ($n = 1$); prior $\mu \sim N(\mu_o, \sigma_o^2)$, $\mu_o = 0$, $\sigma_o^2 = 25$.
- μ_*, σ_*^2: posterior mean and variance; $\eta = 1/\sigma^2$, $\eta_o = 1/\sigma_o^2$, $\eta_* = 1/\sigma_*^2$: precisions.

Step — from scratch. Bayes theorem and expansion of the exponent (dropping terms free of μ):

$$\pi(\mu|x) \propto \exp\Big[-\frac{(x-\mu)^2}{8}\Big] \exp\Big[-\frac{\mu^2}{50}\Big] \tag{Q16.1}$$

$$\propto \exp\Big[-\tfrac{1}{2}\Big\{\Big(\frac{1}{4} + \frac{1}{25}\Big)\mu^2 - 2\mu\frac{x}{4}\Big\}\Big], \tag{Q16.2}$$

so the posterior is normal with precision $\eta_* = 1/4 + 1/25$ and mean $\mu_* = (x/4)/\eta_*$.

Step — numbers (Tsay Result 12.1). Result 12.1 with $n = 1$ gives

$$\mu_* = \frac{\sigma^2\mu_o + n\sigma_o^2 x}{\sigma^2 + n\sigma_o^2} \tag{Q16.3}$$

$$= \frac{(4)(0) + (1)(25)x}{(4) + (1)(25)} = \frac{25}{29}\,x = 0.862069\,x, \tag{Q16.4}$$

$$\sigma_*^2 = \frac{\sigma^2\sigma_o^2}{\sigma^2 + n\sigma_o^2} = \frac{(4)(25)}{(4) + (25)} = \frac{100}{29} = 3.448276. \tag{Q16.5}$$

In precisions, $\eta_* = 0.04 + 0.25 = 0.29$ and $1/0.29 = 3.448276$, matching (Q16.5). Hence

$$\mu|x \sim N\Big(\frac{25}{29}x,\ \frac{100}{29}\Big). \tag{Q16.6}$$

Intuition. The observation is about six times as precise as the prior (0.25 vs. 0.04), so it gets weight $25/29 \approx 0.86$ and the prior mean 0 gets $4/29$; the posterior variance 3.45 is below both 4 and 25.

Ref: Tsay §12.3.2, Result 12.1 (pp. 618–619); Greenberg Exercise 4.2 (p. 58); Question 3 above.

Question 17. Tsay 12.2 — Gibbs conditionals for regression with AR(p) errors.
Source: Tsay, R. S. (2010), *Analysis of Financial Time Series*, 3rd ed., Wiley, Exercise 12.2, p. 670 — reproduced verbatim.

> **12.2.** Consider the linear regression model with time series errors in Section 12.5. Assume that z_t is an AR(p) process (i.e., $z_t = \phi_1 z_{t-1} + \cdots + \phi_p z_{t-p} + a_t$). Let $\boldsymbol{\phi} = (\phi_1, \ldots, \phi_p)'$ be the vector of AR parameters. Derive the conditional posterior distributions of $f(\boldsymbol{\beta}|\boldsymbol{Y}, \boldsymbol{X}, \boldsymbol{\phi}, \sigma^2)$, $f(\boldsymbol{\phi}|\boldsymbol{Y}, \boldsymbol{X}, \boldsymbol{\beta}, \sigma^2)$, and $f(\sigma^2|\boldsymbol{Y}, \boldsymbol{X}, \boldsymbol{\beta}, \boldsymbol{\phi})$ using the conjugate prior distributions, that is, the priors are
>
> $$\boldsymbol{\beta} \sim N(\boldsymbol{\beta}_o, \boldsymbol{\Sigma}_o), \qquad \boldsymbol{\phi} \sim N(\boldsymbol{\phi}_o, \boldsymbol{A}_o), \qquad (v\lambda)/\sigma^2 \sim \chi_v^2.$$
>
> *Context the item refers to (not part of the exercise; Tsay §12.5, p. 624, equation (12.6), reproduced verbatim).*
>
> The model becomes
>
> $$y_t = \boldsymbol{x}_t'\boldsymbol{\beta} + z_t, \qquad z_t = \phi z_{t-1} + a_t, \qquad t = 1, \ldots, n, \tag{12.6}$$
>
> where n is the sample size. [...] $\{a_t\}$ being a sequence of independent and identically distributed normal random variables with mean zero and variance σ^2.

Notation.

- y_t: response; $\boldsymbol{x}_t$: $(k+1) \times 1$ regressors (with a constant); $\boldsymbol{\beta}$: coefficients; $z_t = y_t - \boldsymbol{x}_t'\boldsymbol{\beta}$: regression error; $a_t \sim N(0, \sigma^2)$ i.i.d.
- $\boldsymbol{\phi} = (\phi_1, \ldots, \phi_p)'$; the likelihood conditions on the first p observations and uses $t = p+1, \ldots, n$ (Tsay uses $t = 2, \ldots, n$ for $p = 1$).
- Hyperparameters $\boldsymbol{\beta}_o, \boldsymbol{\Sigma}_o, \boldsymbol{\phi}_o, \boldsymbol{A}_o$ (known), $v, \lambda > 0$; $\sum$ means $\sum_{t=p+1}^{n}$.

Step — conditional likelihood. With $a_t = z_t - \sum_{j=1}^{p} \phi_j z_{t-j}$,

$$f(\boldsymbol{Y}|\boldsymbol{X}, \boldsymbol{\beta}, \boldsymbol{\phi}, \sigma^2) \propto (\sigma^2)^{-(n-p)/2} \exp\Big[-\frac{1}{2\sigma^2}\sum a_t^2\Big]. \tag{Q17.1}$$

Step — (i) $\boldsymbol{\beta}$ given $\boldsymbol{\phi}, \sigma^2$. Define the AR-filtered data

$$y_{o,t} = y_t - \textstyle\sum_{j=1}^{p} \phi_j y_{t-j}, \qquad \boldsymbol{x}_{o,t} = \boldsymbol{x}_t - \textstyle\sum_{j=1}^{p} \phi_j \boldsymbol{x}_{t-j}. \tag{Q17.2}$$

Substituting $z_t = y_t - \boldsymbol{x}_t'\boldsymbol{\beta}$ into a_t:

$$a_t = (y_t - \boldsymbol{x}_t'\boldsymbol{\beta}) - \textstyle\sum_j \phi_j (y_{t-j} - \boldsymbol{x}_{t-j}'\boldsymbol{\beta}) \tag{Q17.3}$$

$$= y_{o,t} - \boldsymbol{x}_{o,t}'\boldsymbol{\beta}, \tag{Q17.4}$$

so $y_{o,t} = \boldsymbol{x}_{o,t}'\boldsymbol{\beta} + a_t$ is a regression with i.i.d. normal errors. Substituting (Q17.4) into (Q17.1) and multiplying by the $N(\boldsymbol{\beta}_o, \boldsymbol{\Sigma}_o)$ prior, the exponent is, up to constants,

$$-\tfrac{1}{2}\Big[\boldsymbol{\beta}'\Big(\frac{\sum \boldsymbol{x}_{o,t}\boldsymbol{x}_{o,t}'}{\sigma^2} + \boldsymbol{\Sigma}_o^{-1}\Big)\boldsymbol{\beta} - 2\boldsymbol{\beta}'\Big(\frac{\sum \boldsymbol{x}_{o,t}y_{o,t}}{\sigma^2} + \boldsymbol{\Sigma}_o^{-1}\boldsymbol{\beta}_o\Big)\Big]. \tag{Q17.5}$$

Completing the square in (Q17.5):

$$\boldsymbol{\beta}|\boldsymbol{Y},\boldsymbol{X},\boldsymbol{\phi},\sigma^2 \sim N(\boldsymbol{\beta}_*,\boldsymbol{\Sigma}_*), \quad \boldsymbol{\Sigma}_*^{-1} = \frac{\sum \boldsymbol{x}_{o,t}\boldsymbol{x}_{o,t}'}{\sigma^2} + \boldsymbol{\Sigma}_o^{-1}, \quad \boldsymbol{\beta}_* = \boldsymbol{\Sigma}_*\Big(\frac{\sum \boldsymbol{x}_{o,t}y_{o,t}}{\sigma^2} + \boldsymbol{\Sigma}_o^{-1}\boldsymbol{\beta}_o\Big), \tag{Q17.6}$$

which is Tsay's (12.9) with p lags in the filter.

Step — (ii) $\boldsymbol{\phi}$ given $\boldsymbol{\beta}, \sigma^2$. With $\boldsymbol{\beta}$ known, $z_t = y_t - \boldsymbol{x}_t'\boldsymbol{\beta}$ is observed. Let $\boldsymbol{w}_t = (z_{t-1}, \ldots, z_{t-p})'$:

$$z_t = \boldsymbol{w}_t'\boldsymbol{\phi} + a_t, \qquad t = p+1, \ldots, n, \tag{Q17.7}$$

again a normal regression. The same completing of the square with the $N(\boldsymbol{\phi}_o, \boldsymbol{A}_o)$ prior gives

$$\boldsymbol{\phi}|\boldsymbol{Y},\boldsymbol{X},\boldsymbol{\beta},\sigma^2 \sim N(\boldsymbol{\phi}_*,\boldsymbol{A}_*), \quad \boldsymbol{A}_*^{-1} = \frac{\sum \boldsymbol{w}_t\boldsymbol{w}_t'}{\sigma^2} + \boldsymbol{A}_o^{-1}, \quad \boldsymbol{\phi}_* = \boldsymbol{A}_*\Big(\frac{\sum \boldsymbol{w}_t z_t}{\sigma^2} + \boldsymbol{A}_o^{-1}\boldsymbol{\phi}_o\Big), \tag{Q17.8}$$

the AR(p) version of (12.10). (If stationarity is imposed, (Q17.8) is truncated to the stationary region, e.g. by rejecting draws outside it.)

Step — (iii) σ^2 given $\boldsymbol{\beta}, \boldsymbol{\phi}$. The prior $v\lambda/\sigma^2 \sim \chi^2_v$ is an inverted chi-squared, with density

$$\pi(\sigma^2) \propto (\sigma^2)^{-(v/2+1)} \exp\Big[-\frac{v\lambda}{2\sigma^2}\Big]. \tag{Q17.9}$$

Multiplying (Q17.1) by (Q17.9), with a_t computed from $\boldsymbol{\beta}$ and $\boldsymbol{\phi}$:

$$\pi(\sigma^2|\boldsymbol{Y},\boldsymbol{X},\boldsymbol{\beta},\boldsymbol{\phi}) \propto (\sigma^2)^{-((v+n-p)/2+1)} \exp\Big[-\frac{v\lambda + \sum a_t^2}{2\sigma^2}\Big], \tag{Q17.10}$$

$$\frac{v\lambda + \sum_{t=p+1}^{n} a_t^2}{\sigma^2} \sim \chi^2_{v+n-p}, \tag{Q17.11}$$

which is (12.11) for $p = 1$.

Step — numerical check (seed 122; function `t122`). Simulated data: $n = 500$, $\boldsymbol{x}_t = (1, x_t)'$ with $x_t \sim N(0,1)$, $\boldsymbol{\beta} = (1,2)'$, $\boldsymbol{\phi} = (0.5, -0.3)'$, $\sigma^2 = 1$; priors $\boldsymbol{\beta}_o = \mathbf{0}$, $\boldsymbol{\Sigma}_o = 100\boldsymbol{I}$, $\boldsymbol{\phi}_o = \mathbf{0}$, $\boldsymbol{A}_o = \boldsymbol{I}$, $v = 3$, $\lambda = 1$. Cycling (Q17.6), (Q17.8), (Q17.11) for 2,100 iterations and discarding 100 (as in Tsay's Example 12.1):

```
yo = y[p:] - lagmat(y) @ phi                                    # (Q17.2)
Xo = X[p:] - sum(phi[j] * X[p-j-1:n-j-1] for j in range(p))
Sst = np.linalg.inv(Xo.T @ Xo / s2 + Sinv)
beta = Sst @ (Xo.T @ yo / s2 + Sinv @ bo) + np.linalg.cholesky(Sst) @ rng.normal(size=2)
zz = y - X @ beta; W, zt = lagmat(zz), zz[p:]
Ast = np.linalg.inv(W.T @ W / s2 + Ainv)
phi = Ast @ (W.T @ zt / s2 + Ainv @ phio) + np.linalg.cholesky(Ast) @ rng.normal(size=p)
at = zt - W @ phi
s2 = (v * lam + at @ at) / rng.chisquare(v + n - p)           # (Q17.11)
```

Posterior means (s.d.): $\beta_0 = 1.0317$ (0.0550), $\beta_1 = 2.0554$ (0.0394), $\phi_1 = 0.5182$ (0.0415), $\phi_2 = -0.3652$ (0.0413), $\sigma^2 = 1.0533$ (0.0669); every true value is within two posterior s.d.

Intuition. Each block is a textbook conjugate regression once the other blocks are fixed: knowing $\boldsymbol{\phi}$ whitens the errors (Cochrane–Orcutt filtering), knowing $\boldsymbol{\beta}$ reveals the error series, and knowing both reveals the innovations.

Ref: Tsay §12.5 (pp. 624–628), Results 12.1a and 12.8 (pp. 619–621); Greenberg §10.1 (AR models).

Question 18. Tsay 12.3 — two consecutive missing values in an AR(p).
Source: Tsay, R. S. (2010), *Analysis of Financial Time Series*, 3rd ed., Wiley, Exercise 12.3, p. 670 — reproduced verbatim.

> **12.3.** Consider the linear AR(p) model in Section 12.6.1. Suppose that x_h and x_{h+1} are two missing values with a joint prior distribution being multivariate normal with mean $\boldsymbol{\mu}_o$ and covariance matrix $\boldsymbol{\Sigma}_o$. Other prior distributions are the same as that in the text. What is the conditional posterior distribution of the two missing values?
>
> *Context the item refers to (not part of the exercise; Tsay §12.6.1, pp. 629–630, reproduced verbatim).*
>
> For ease in presentation, consider an AR(p) time series
>
> $$x_t = \phi_1 x_{t-1} + \cdots + \phi_p x_{t-p} + a_t, \tag{12.15}$$
>
> where $\{a_t\}$ is a Gaussian white noise series with mean zero and variance σ^2. [...] If we assume that the prior distributions are
>
> $$\boldsymbol{\phi} \sim N(\boldsymbol{\phi}_o, \boldsymbol{\Sigma}_o), \qquad x_h \sim N(\mu_o, \sigma_o^2), \qquad \frac{v\lambda}{\sigma^2} \sim \chi^2_v,$$
>
> where the hyperparameters are known [...].

Notation.

- x_t: AR(p) series (12.15), $t = 1, \ldots, n$; $\boldsymbol{m} = (x_h, x_{h+1})'$: the two missing values, with $h - p \geq 1$ and $h + 1 + p \leq n$.
- $\psi_0 = 1$, $\psi_j = -\phi_j$ $(j = 1, \ldots, p)$, so $a_t = \sum_{j=0}^{p} \psi_j x_{t-j}$; $\psi_j \equiv 0$ outside $0 \leq j \leq p$.
- $\boldsymbol{\mu}_o$, $\boldsymbol{\Sigma}_o$: prior mean and covariance of $\boldsymbol{m}$; $\boldsymbol{\mu}_*$, $\boldsymbol{\Sigma}_*$: posterior ones.

Step — which equations involve $\boldsymbol{m}$. x_h enters a_t for $t = h, \ldots, h + p$ and x_{h+1} for $t = h + 1, \ldots, h + 1 + p$: the $p + 2$ equations $t = h, \ldots, h + p + 1$. Splitting each into the missing and the observed parts:

$$a_t = c_{t1} x_h + c_{t2} x_{h+1} + \textstyle\sum_{j:\, t-j \notin \{h, h+1\}} \psi_j x_{t-j}, \tag{Q18.1}$$

$$c_{t1} = \psi_{t-h}, \qquad c_{t2} = \psi_{t-h-1}. \tag{Q18.2}$$

Define the observed quantity $y_t^* = -\sum_{j:\, t-j \notin \{h, h+1\}} \psi_j x_{t-j}$ and $b_t = -a_t \sim N(0, \sigma^2)$ (the normal is symmetric, as in Tsay). Then (Q18.1) becomes

$$y_t^* = \boldsymbol{c}_t' \boldsymbol{m} + b_t, \qquad t = h, \ldots, h + p + 1, \tag{Q18.3}$$

$$\boldsymbol{y}^* = \boldsymbol{C}\boldsymbol{m} + \boldsymbol{b}, \qquad \boldsymbol{C} = (\boldsymbol{c}_h, \ldots, \boldsymbol{c}_{h+p+1})' \ \ ((p+2) \times 2), \tag{Q18.4}$$

a linear regression with two unknown "coefficients" x_h, x_{h+1}. No other equation of the likelihood involves $\boldsymbol{m}$.

Step — posterior. Given $\boldsymbol{\phi}$, σ^2 and the observed data, multiply the likelihood of (Q18.4) by the prior of $\boldsymbol{m}$:

$$\pi(\boldsymbol{m} | \boldsymbol{X}, \boldsymbol{\phi}, \sigma^2) \propto \exp\Big[-\frac{(\boldsymbol{y}^* - \boldsymbol{C}\boldsymbol{m})'(\boldsymbol{y}^* - \boldsymbol{C}\boldsymbol{m})}{2\sigma^2} - \frac{(\boldsymbol{m} - \boldsymbol{\mu}_o)' \boldsymbol{\Sigma}_o^{-1} (\boldsymbol{m} - \boldsymbol{\mu}_o)}{2} \Big]. \tag{Q18.5}$$

Completing the square in $\boldsymbol{m}$ (Result 12.1a with the least-squares estimate $\hat{\boldsymbol{m}} = (\boldsymbol{C}'\boldsymbol{C})^{-1}\boldsymbol{C}'\boldsymbol{y}^*$, $\mathrm{Var}(\hat{\boldsymbol{m}}) = \sigma^2(\boldsymbol{C}'\boldsymbol{C})^{-1}$):

$$(x_h, x_{h+1})' \,|\, \boldsymbol{X}, \boldsymbol{\phi}, \sigma^2 \sim N_2(\boldsymbol{\mu}_*, \boldsymbol{\Sigma}_*), \tag{Q18.6}$$

$$\boldsymbol{\Sigma}_*^{-1} = \boldsymbol{\Sigma}_o^{-1} + \frac{\boldsymbol{C}'\boldsymbol{C}}{\sigma^2}, \tag{Q18.7}$$

$$\boldsymbol{\mu}_* = \boldsymbol{\Sigma}_*\Big(\boldsymbol{\Sigma}_o^{-1}\boldsymbol{\mu}_o + \frac{\boldsymbol{C}'\boldsymbol{y}^*}{\sigma^2}\Big). \tag{Q18.8}$$

Step — the entries of $\boldsymbol{C}'\boldsymbol{C}$. From (Q18.2), as t runs over the $p+2$ equations each column of $\boldsymbol{C}$ runs over $\psi_0, \dots, \psi_p$, shifted by one row:

$$(\boldsymbol{C}'\boldsymbol{C})_{11} = (\boldsymbol{C}'\boldsymbol{C})_{22} = \textstyle\sum_{j=0}^{p} \psi_j^2 = 1 + \sum_{j=1}^{p} \phi_j^2, \tag{Q18.9}$$

$$(\boldsymbol{C}'\boldsymbol{C})_{12} = \textstyle\sum_{j=1}^{p} \psi_j \psi_{j-1} = -\phi_1 + \sum_{j=2}^{p} \phi_j \phi_{j-1}. \tag{Q18.10}$$

For $p=1$: $\boldsymbol{c}_h = (1,0)'$, $\boldsymbol{c}_{h+1} = (-\phi, 1)'$, $\boldsymbol{c}_{h+2} = (0, -\phi)'$; $y^*_h = \phi x_{h-1}$, $y^*_{h+1} = 0$, $y^*_{h+2} = -x_{h+2}$; so

$$\boldsymbol{C}'\boldsymbol{C} = \begin{pmatrix} 1+\phi^2 & -\phi \\ -\phi & 1+\phi^2 \end{pmatrix}, \qquad \boldsymbol{C}'\boldsymbol{y}^* = \begin{pmatrix} \phi x_{h-1} \\ \phi x_{h+2} \end{pmatrix}. \tag{Q18.11}$$

The script (function `t123`) builds $\boldsymbol{C}$ for an AR(2), checks (Q18.9)–(Q18.10), and verifies at random points that the log of prior × full AR likelihood differs from its value at $\boldsymbol{\mu}_*$ by exactly $-\frac{1}{2}(\boldsymbol{m} - \boldsymbol{\mu}_*)'\boldsymbol{\Sigma}_*^{-1}(\boldsymbol{m} - \boldsymbol{\mu}_*)$ (to 10^{-9}).

Intuition. The two gaps are filled from both sides: p values before and p after inform them, through the AR equations, as a small regression. The off-diagonal $-\phi_1/\sigma^2$ (for $p=1$) in the precision makes adjacent missing values positively correlated when $\phi_1 > 0$, which is why Tsay recommends drawing them jointly in a Gibbs sampler.

Ref: Tsay §12.6.1 (pp. 629–631), eqs. (12.16)–(12.17) for one missing value; Result 12.1a (p. 619).

Block 3 — Bolfarine & Sandoval (2010) *(Chs. 4–5 • 5 questions)*

Question 19. Bolfarine & Sandoval 4.3 — risk and Bayes risk, binomial.
Source: Bolfarine, H. and Sandoval, M. C. (2010), *Introdução à Inferência Estatística*, 2nd ed., SBM, Exercício 4.3, p. 71 — reproduced verbatim (Portuguese original).

> **4.3.** Considere uma única observação da variável aleatória $X \sim Binomial(m, \theta)$. Seja $l(\theta, d) = (\theta - d)^2$.
> (i) Encontre o risco de $d(X) = X/m$.
> (ii) Encontre o risco de Bayes de $d(X)$ em (i), com relação a priori $\pi(\theta) = 1, 0 \le \theta \le 1$.

Notation.
- $X \sim \text{Binomial}(m, \theta)$, $m \ge 1$, $\theta \in [0,1]$; $d(X) = X/m$; loss $l(\theta, d) = (\theta - d)^2$.
- $R(\theta, d) = E_\theta[l(\theta, d(X))]$: risk; $r(\pi, d) = E_\pi[R(\theta, d)]$: Bayes risk under the uniform prior.

Step — (i) risk. Squared-error risk is variance plus squared bias, and $E(X) = m\theta$, $\text{Var}(X) = m\theta(1-\theta)$:

$$R(\theta, d) = E\big[(\theta - X/m)^2\big] = \text{Var}(X/m) + \big(E[X/m] - \theta\big)^2 \tag{Q19.1}$$

$$= \frac{m\theta(1-\theta)}{m^2} + \Big(\frac{m\theta}{m} - \theta\Big)^2 \tag{Q19.2}$$

$$= \frac{\theta(1-\theta)}{m}. \tag{Q19.3}$$

Step — (ii) Bayes risk. Averaging (Q19.3) over $\pi(\theta) = 1$:

$$r(\pi, d) = \int_0^1 \frac{\theta(1-\theta)}{m}\,(1)\,d\theta \tag{Q19.4}$$

$$= \frac{1}{m}\Big(\frac{1}{2} - \frac{1}{3}\Big) \tag{Q19.5}$$

$$= \frac{1}{6m}. \tag{Q19.6}$$

(The script checks (Q19.3) by summing over the binomial p.m.f. and (Q19.6) by quadrature, for $m = 10$.)

Intuition. X/m is unbiased, so its risk is all variance, largest at $\theta = 1/2$. The Bayes estimator under this prior, $(X + 1)/(m + 2)$, has the smaller Bayes risk $1/[6(m + 2)]$ (Bolfarine & Sandoval Example 4.4.1): a little bias buys a larger cut in variance.

Ref: Bolfarine & Sandoval §4.1–4.4 (pp. 55–70), Example 4.4.1.

Question 20. Bolfarine & Sandoval 4.5 — Bayes risk of X under a gamma prior.
Source: Bolfarine, H. and Sandoval, M. C. (2010), *Introdução à Inferência Estatística*, 2nd ed., SBM, Exercício 4.5, p. 71 — reproduced verbatim (Portuguese original).

> **4.5.** Seja uma única observação da distribuição $Poisson(\theta)$. Encontre o risco de Bayes do estimador $d(X) = X$, com relação à perda quadrática e a priori $Gama(\alpha, \beta)$.

Notation.
- $X \sim \text{Poisson}(\theta)$, one observation; $d(X) = X$; quadratic loss $(\theta - d)^2$.
- Prior $Gama(\alpha, \beta)$: $\pi(\theta) = \beta^\alpha \theta^{\alpha-1} e^{-\beta\theta}/\Gamma(\alpha)$ (Bolfarine & Sandoval Example 2.3.4), $\alpha, \beta > 0$.

Step — risk. $d(X) = X$ is unbiased ($E_\theta X = \theta$), so

$$R(\theta, d) = E_\theta(\theta - X)^2 = \text{Var}_\theta(X) = \theta. \tag{Q20.1}$$

Step — Bayes risk. Averaging (Q20.1) over the prior and using the gamma integral (Q5.3):

$$r(\pi, d) = \int_0^\infty \theta \, \frac{\beta^\alpha}{\Gamma(\alpha)} \theta^{\alpha-1} e^{-\beta\theta} d\theta \tag{Q20.2}$$

$$= \frac{\beta^\alpha}{\Gamma(\alpha)} \cdot \frac{\Gamma(\alpha + 1)}{\beta^{\alpha+1}} \tag{Q20.3}$$

$$= \frac{\alpha}{\beta}. \tag{Q20.4}$$

Step — comparison. The Bayes estimator under this prior is the posterior mean $(\alpha + X)/(\beta + 1)$, whose Bayes risk is the expected posterior variance:

$$E\Big[\frac{\alpha + X}{(\beta + 1)^2}\Big] = \frac{\alpha + \alpha/\beta}{(\beta + 1)^2} = \frac{\alpha}{\beta(\beta + 1)} < \frac{\alpha}{\beta}, \tag{Q20.5}$$

using $E(X) = E[E(X|\theta)] = E(\theta) = \alpha/\beta$ (checked numerically in the script).

Intuition. The Bayes risk of X is simply the prior mean of its variance θ; shrinking X toward the prior mean divides that risk by $\beta + 1$.

Ref: Bolfarine & Sandoval §4.3 (Definition 4.3.1, p. 61), Example 4.4.2 (pp. 67–68).

Question 21. Bolfarine & Sandoval 4.9 — Bayes estimator of $P_\theta(X = 0)$.
Source: Bolfarine, H. and Sandoval, M. C. (2010), *Introdução à Inferência Estatística*, 2nd ed., SBM, Exercício 4.9, p. 72 — reproduced verbatim (Portuguese original).

> **4.9.** Em uma área de reflorestamento, o número de árvores de determinada espécie, por hectare, com certa doença tem uma distribuição $Poisson(\theta)$. A distribuição a priori de θ é exponencial com média igual a 1. Encontre o estimador de Bayes de $P_\theta(X = 0)$ com relação à perda quadrática..

Notation.
- $X_1, \ldots, X_n$: diseased trees in n hectares, i.i.d. Poisson(θ) (the statement's single hectare is $n = 1$); $s = \sum_i X_i$.
- Prior: exponential with mean 1, $\pi(\theta) = e^{-\theta}$ $(= Gama(1, 1))$; $g(\theta) = P_\theta(X = 0) = e^{-\theta}$.

Step — posterior. As in (Q1.6) of Question 1 with $\alpha = \beta = 1$:

$$\pi(\theta|x) \propto \left(\theta^s e^{-n\theta}\right)\left(e^{-\theta}\right) = \theta^s e^{-(n+1)\theta}, \tag{Q21.1}$$

i.e. $\theta|x \sim Gama(s+1, n+1)$.

Step — Bayes estimator. Under quadratic loss the Bayes estimator of $g(\theta)$ is its posterior mean (Theorem 4.4.1 applied to $g(\theta)$):

$$d_B(x) = E[e^{-\theta}|x] = \int_0^\infty e^{-\theta}\,\frac{(n+1)^{s+1}}{\Gamma(s+1)}\theta^s e^{-(n+1)\theta}d\theta \tag{Q21.2}$$

$$= \frac{(n+1)^{s+1}}{\Gamma(s+1)}\int_0^\infty \theta^s e^{-(n+2)\theta}d\theta. \tag{Q21.3}$$

Substituting the gamma integral (Q5.3) with $a = s+1$, $b = n+2$:

$$d_B(x) = \frac{(n+1)^{s+1}}{\Gamma(s+1)}\cdot\frac{\Gamma(s+1)}{(n+2)^{s+1}} \tag{Q21.4}$$

$$= \left(\frac{n+1}{n+2}\right)^{s+1}. \tag{Q21.5}$$

For one hectare ($n = 1$, $X = x$):

$$d_B(x) = \left(\frac{2}{3}\right)^{x+1}, \tag{Q21.6}$$

e.g. $2/3$ for $x = 0$ and $(2/3)^4 = 0.1975$ for $x = 3$.

Intuition. d_B is not $e^{-E(\theta|x)}$ (by Jensen's inequality $E[e^{-\theta}|x] > e^{-E[\theta|x]}$), and unlike the plug-in $e^{-\bar{x}}$ it never claims certainty: after seeing $x = 0$ the plug-in says $P(X = 0) = 1$, the Bayes estimator says $2/3$.

Ref: Bolfarine & Sandoval §4.4, Theorem 4.4.1 (p. 63), Example 4.4.2.

Question 22. Bolfarine & Sandoval 4.10 — Pareto prior for $U(0, \theta)$.

Source: Bolfarine, H. and Sandoval, M. C. (2010), *Introdução à Inferência Estatística*, 2nd ed., SBM, Exercício 4.10, p. 72 — reproduced verbatim (Portuguese original).

> **4.10.** Sejam $X_1, \ldots, X_n$ uma amostra aleatória da distribuição $U(0, \theta)$. Suponhamos que θ seja uma variável aleatória com função de densidade de probabilidade (Pareto)
>
> $$\pi(\theta) = \begin{cases} ba^b/\theta^{b+1}, & \theta \geq a, \\ 0, & \theta < a, \end{cases}$$
>
> Encontre a distribuição a posteriori de θ e o estimador de Bayes de θ com relação à perda quadrática.

Notation.

- $X_i \sim U(0, \theta)$, $i = 1, \ldots, n$; $X_{(n)} = \max_i X_i$; I_A: indicator.
- Pareto prior: scale $a > 0$ (lower bound), shape $b > 0$ (Greenberg's k and a in Question 2); $M = \max(a, X_{(n)})$.

Step — posterior. By (Q2.2) of Question 2 the likelihood is $\theta^{-n} I(\theta \geq X_{(n)})$; times the prior,

$$\pi(\theta|\boldsymbol{X}) \propto \left(\theta^{-n} I(\theta \geq X_{(n)})\right)\left(ba^b\theta^{-(b+1)} I(\theta \geq a)\right) \tag{Q22.1}$$

$$\propto \theta^{-(n+b+1)} I(\theta \geq M). \tag{Q22.2}$$

Normalising with $\int_M^\infty \theta^{-(n+b+1)}d\theta = M^{-(n+b)}/(n+b)$:

$$h(\theta|\boldsymbol{X}) = \frac{(n+b)\,M^{n+b}}{\theta^{n+b+1}}\,I_{[M,\infty)}(\theta), \qquad M = \max(a, X_{(n)}), \tag{Q22.3}$$

a Pareto with shape $n+b$ and scale M (this is the book's (5.5.2), quoted in Example 5.5.2).

Step — Bayes estimator. Under quadratic loss it is the posterior mean (Theorem 4.4.1):

$$d_B(\boldsymbol{X}) = \int_M^\infty \theta\,\frac{(n+b)M^{n+b}}{\theta^{n+b+1}}d\theta = (n+b)M^{n+b}\,\frac{M^{-(n+b-1)}}{n+b-1} \tag{Q22.4}$$

$$= \frac{(n+b)\max(a, X_{(n)})}{n+b-1}. \tag{Q22.5}$$

Intuition. Same structure as Question 2: the data move the lower bound to $X_{(n)}$, and the estimator inflates the bound by $(n+b)/(n+b-1)$; a large b (a prior concentrated near a) or a large n brings the factor close to one.

Ref: Bolfarine & Sandoval §4.4 (Theorem 4.4.1), Example 5.5.2 (pp. 86–87); Greenberg Exercise 2.4.

Question 23. Bolfarine & Sandoval 5.11 — equal-tailed Bayesian interval for a Poisson mean.
Source: Bolfarine, H. and Sandoval, M. C. (2010), *Introdução à Inferência Estatística*, 2nd ed., SBM, Exercício 5.11, p. 89 — reproduced verbatim (Portuguese original).

> **5.11.** Sejam $X_1, \ldots, X_n$ uma amostra aleatória de tamanho n da distribuição $Poisson(\theta)$, com priori
>
> $$\pi(\theta) = e^{-\theta}, \quad \theta > 0.$$
>
> Construa um intervalo de confiança Bayesiano simétrico para θ com $\gamma = 0{,}95$. Se $n = 10$ e $\sum_{i=1}^n X_i = 18$, como fica o intervalo?

Notation.

- $X_i \sim \text{Poisson}(\theta)$, $s = \sum_i X_i$; prior $e^{-\theta} = Gama(1,1)$; $\gamma = 0.95$ (the book's 0,95), $\alpha = 1-\gamma$.
- "Symmetric" interval (Bolfarine & Sandoval §5.5): equal posterior probability $\alpha/2$ in each tail; $\chi^2_{\nu;q}$: q-quantile of χ^2_ν.

Step — posterior. By (Q21.1) of Question 21, $\theta|\boldsymbol{X} \sim Gama(s+1, n+1)$.

Step — pivot to χ^2. If $\theta \sim Gama(a,b)$ (rate b), then $b\theta \sim Gama(a,1)$ and $2b\theta \sim Gama(a,1/2) = \chi^2_{2a}$. With $a = s+1$, $b = n+1$:

$$2(n+1)\theta\,|\,\boldsymbol{X} \sim \chi^2_{2(s+1)}. \tag{Q23.1}$$

Hence, with $q_1 = \chi^2_{2(s+1);\alpha/2}$ and $q_2 = \chi^2_{2(s+1);1-\alpha/2}$, $P(q_1 \le 2(n+1)\theta \le q_2|\boldsymbol{X}) = \gamma$, and

$$\Big[\frac{q_1}{2(n+1)};\ \frac{q_2}{2(n+1)}\Big] \tag{Q23.2}$$

is the symmetric Bayesian interval.

Step — numbers. $n = 10$, $s = 18$: $2(s+1) = 38$, $2(n+1) = 22$, $q_1 = \chi^2_{38;0.025} = 22.8785$, $q_2 = \chi^2_{38;0.975} = 56.8955$. Substituting into (Q23.2):

$$\Big[\frac{22.8785}{22};\ \frac{56.8955}{22}\Big] = [1.0399;\ 2.5862]. \tag{Q23.3}$$

The posterior mean is $19/11 = 1.7273$ (MLE $\bar{x} = 1.8$).

Intuition. Given the data, θ lies in $[1.04, 2.59]$ with probability 0.95, a direct probability statement about θ. Because the gamma posterior is right-skewed, this equal-tailed interval is not the shortest (HPD) one.

Ref: Bolfarine & Sandoval §5.5 (pp. 85–87), Definition 5.5.1; Greenberg §3.2.2 (interval estimates).

Block 4 — Morettin & Singer (2021) *(Apps. B–C • 3 questions)*

Question 24. Morettin & Singer B.16 — accept–reject for a truncated gamma.
Source: Morettin, P. A. and Singer, J. M. (2021), *Estatística e Ciência de Dados* (preliminary version, April 2021), Apêndice B, Exercício 16, p. 478 — reproduced verbatim (Portuguese original).

> 16) Considere a distribuição gama truncada $\pi(x) \propto x^2 \exp(-x)$, $0 \le x \le 1$. Simule uma amostra de tamanho 1.000 dessa distribuição usando o método da aceitação-rejeição com $g(x) = \exp(-x)$.

Notation.
- Target $\pi(x) = x^2 e^{-x}/Z$ on $[0,1]$, $Z = \int_0^1 x^2 e^{-x} dx$; proposal $g(x) = e^{-x}$, $x > 0$ (Exp(1)), as the statement says.
- c: constant with $\pi \le cg$; $N = 1{,}000$ accepted draws.

Step — bound and acceptance rule. On $[0,1]$ the ratio is

$$\frac{\pi(x)}{g(x)} = \frac{x^2}{Z}\,1(0 \le x \le 1) \le \frac{1}{Z} = c, \tag{Q24.1}$$

$$\frac{\pi(x)}{c\,g(x)} = x^2\,1(x \le 1), \tag{Q24.2}$$

so a proposal $x \sim \text{Exp}(1)$ is accepted with probability x^2 if $x \le 1$ and always rejected if $x > 1$. The overall acceptance probability is $1/c = Z$.

Step — Z and the true moments. Integrating by parts, $\int_0^1 x^2 e^{-x} dx = [-e^{-x}(x^2 + 2x + 2)]_0^1$:

$$Z = 2 - 5e^{-1} = 2 - 1.839397 = 0.160603, \tag{Q24.3}$$

$$E(x) = \frac{\int_0^1 x^3 e^{-x} dx}{Z} = \frac{6 - 16e^{-1}}{Z} = \frac{6 - 5.886071}{0.160603} = \frac{0.113929}{0.160603} = 0.709383, \tag{Q24.4}$$

$$E(x^2) = \frac{24 - 65e^{-1}}{Z} = \frac{24 - 23.912164}{0.160603} = \frac{0.087836}{0.160603} = 0.546917, \tag{Q24.5}$$

$$\text{Var}(x) = 0.546917 - (0.709383)^2 = 0.546917 - 0.503225 = 0.0437. \tag{Q24.6}$$

Step — code (seed 16; function `ms_b16`).

```
acc, tried = [], 0
while len(acc) < 1000:
    xg = rng.exponential(1.0)                  # draw from g(x) = exp(-x)
    tried += 1
    if xg <= 1 and rng.uniform() <= xg ** 2:   # accept with prob x^2 1(x <= 1)  (Q24.2)
        acc.append(xg)
```

Output: 6,644 proposals for 1,000 acceptances (rate 0.1505); sample mean 0.7105, sample variance 0.0443, against 0.7094 and 0.0437 from (Q24.4) and (Q24.6).

Step — comparison. The number of proposals needed is negative binomial with mean $N/Z = 6{,}226.5$ and s.d. $\sqrt{N(1-Z)}/Z = 180.4$; 6,644 is 2.3 s.d. above the mean, an unlucky but legitimate run. The moments of the accepted draws agree with the truth (mean error 0.0011, well inside $3\sqrt{0.0437/1000}$).

Intuition. The statement's proposal wastes most draws: 37% fall above 1 and are rejected outright, and small x are accepted with probability x^2. Restricting g to $[0,1]$ (density $e^{-x}/(1-e^{-1})$) would raise the acceptance rate to $Z/(1-e^{-1}) = 0.2541$ with the same rule.

Ref: Morettin & Singer §B.6.1 (amostragem por aceitação-rejeição, pp. 468–470); Greenberg §5.3.

Question 25. Morettin & Singer B.19 — accept–reject for discrete variables.
Source: Morettin, P. A. and Singer, J. M. (2021), *Estatística e Ciência de Dados* (preliminary version, April 2021), Apêndice B, Exercício 19, p. 478 — reproduced verbatim (Portuguese original).

> 19) Considere o método da aceitação-rejeição no caso de variáveis discretas. Especificamente, suponha que queiramos simular valores de uma variável aleatória X, com probabilidades $\pi_j = P(X = j)$ e dispomos de um método para simular valores da variável aleatória Y, com probabilidades $g_j = P(Y = j)$. Suponha ainda, que exista uma constante c, tal que $\pi_j \leq cg_j$, para todo j. Usando o algoritmo de aceitação-rejeição, mostre que $P(\text{aceitar}) = 1/c$ e que o número de iterações necessárias para obter X é uma variável aleatória com distribuição geométrica com média c. Com esse resultado, mostre que $P(X = j) = \pi_j$.

Notation.

- π_j: target p.m.f.; g_j: proposal p.m.f. with $g_j > 0$ wherever $\pi_j > 0$; $c \geq 1$ with $\pi_j \leq cg_j$ for all j.
- Each iteration: draw $Y \sim g$ and $U \sim U(0,1)$ independently; accept $X = Y$ if $U \leq \pi_Y/(cg_Y)$. N: number of iterations until the first acceptance.

Step — $P(\textbf{accept}) = 1/c$. Conditioning on Y (the ratio $\pi_j/(cg_j) \leq 1$ is a valid probability):

$$P(\text{accept}) = \sum_j P(Y = j)\, P\Big(U \leq \frac{\pi_j}{cg_j}\Big) \tag{Q25.1}$$

$$= \sum_j g_j \, \frac{\pi_j}{cg_j} \tag{Q25.2}$$

$$= \frac{1}{c} \sum_j \pi_j = \frac{1}{c}. \tag{Q25.3}$$

Step — N is geometric with mean c. Iterations are independent and each succeeds with probability $1/c$ by (Q25.3), so

$$P(N = k) = \Big(1 - \frac{1}{c}\Big)^{k-1} \frac{1}{c}, \qquad k = 1, 2, \ldots, \tag{Q25.4}$$

$$E(N) = \frac{1}{1/c} = c. \tag{Q25.5}$$

Step — $P(X = j) = \pi_j$. In a single iteration,

$$P(Y = j,\ \text{accept}) = g_j \, \frac{\pi_j}{cg_j} = \frac{\pi_j}{c}. \tag{Q25.6}$$

$X = j$ means the first $k - 1$ iterations reject and the k-th accepts with $Y = j$, for some k:

$$P(X = j) = \sum_{k=1}^{\infty} \Big(1 - \frac{1}{c}\Big)^{k-1} \frac{\pi_j}{c} \tag{Q25.7}$$

$$= \frac{\pi_j}{c} \cdot \frac{1}{1 - (1 - 1/c)} = \frac{\pi_j}{c} \cdot c = \pi_j. \tag{Q25.8}$$

Step — illustration (added, not in B.19: the numbers of Exercise B.20). $\pi = (0.2, 0.3, 0.1, 0.1, 0.3)$ on $\{1, \ldots, 5\}$ and $g_j = 1/5$: the smallest valid constant is $c = \max_j \pi_j/g_j = 0.3/0.2 = 1.5$, so $P(\text{accept}) = 2/3$ and $E(N) = 1.5$. Simulating 10,000 values (seed 19; function `ms_b19`) gives mean $N = 1.5026$ and frequencies $(0.1993, 0.3045, 0.0985, 0.0995, 0.2982)$.

Intuition. Acceptance re-weights the proposal by π_j/g_j; the constant c only sets how often we must retry, never the shape of the output. That is why the smallest admissible $c = \max_j \pi_j/g_j$ is the best choice.

Ref: Morettin & Singer §B.6.1 (pp. 468–470, eq. (B.14)); Greenberg §5.3 (continuous proof, p. 67).

Question 26. Morettin & Singer C.5 — chained data (data augmentation) algorithm.
Source: Morettin, P. A. and Singer, J. M. (2021), *Estatística e Ciência de Dados* (preliminary version, April 2021), Apêndice C, Exercício 5, p. 489 — reproduced verbatim (Portuguese original).

> 5) **Algoritmo de Dados Encadeados** (*Chain Data Algorithm*)
> Numa abordagem um pouco diferente do algoritmos para dados aumentados, iteramos sucessivamente de $p(\mathbf{z}|\boldsymbol{\theta},\mathbf{y})$ e $p(\boldsymbol{\theta}|\mathbf{y},\mathbf{z})$ para obter uma sequência de realizações $(\mathbf{z}^{(1)},\boldsymbol{\theta}^{(1)}),\ldots,(\mathbf{z}^{(n)},\boldsymbol{\theta}^{(n)})$ com as simulações de uma variável realizadas condicionalmente aos valores atuais da outra variável. Esperamos que a distribuição de equilíbrio dessa cadeia seja a distribuição conjunta $p(\mathbf{z},\boldsymbol{\theta}|\mathbf{y})$. Este algoritmo é um caso especial do amostrador de Gibbs, descrito na Nota de Capítulo 2 do Apêndice B. Normalmente, estaremos interessados nas distribuições marginais.
>
> Para os dados do Exemplo C.2, obtenha as sequências de realizações de Z e θ. Simule $n = 200$ valores, desprezando os primeiros 50 valores, conhecidos como **valores de aquecimento**) (*burn-in*).

Context the item refers to (not part of the exercise; Morettin & Singer, Exemplo C.2, p. 480, excerpt, reproduced verbatim).

> **Exemplo C.2.** Retomemos o problema da ligação genética discutido no Exemplo B.16, segundo o qual os dados $\mathbf{y} = (y_1, y_2, y_3, y_4)^\top = (125, 18, 20, 34)^\top$ ocorrem com probabilidades $\pi_1 = (2+\theta)/4$, $\pi_2 = \pi_3 = (1-\theta)/4$, $\pi_4 = \theta/4$, $0 < \theta < 1$. [...] Admitamos que $\mathbf{x} = (x_1, \ldots, x_5)^\top$ sejam os dados completos que ocorrem, respectivamente, com probabilidades $(1/2, \theta/4, (1-\theta)/4, (1-\theta)/4, \theta/4)$, mas que só observamos $y_1 = x_1 + x_2$, $y_2 = x_3$, $y_3 = x_4$ e $y_4 = x_5$, [...] Podemos simplificar as expressões acima obtendo
>
> $$L(\theta|\mathbf{y}) \propto (2+\theta)^{y_1}(1-\theta)^{y_2+y_3}\theta^{y_4}, \tag{C.2}$$
>
> e
>
> $$L(\theta|\mathbf{X}) \propto \theta^{x_2+x_5}(1-\theta)^{x_3+x_4}. \tag{C.3}$$

Notation.

- $\mathbf{y} = (125, 18, 20, 34)$; $\theta \in (0,1)$; latent $Z = x_2$ (the part of $y_1 = 125$ in the cell of probability $\theta/4$), so $x_3 = 18$, $x_4 = 20$, $x_5 = 34$.
- Prior: $\theta \sim U(0,1)$ (flat, so the posterior is proportional to the likelihood (C.2), the target of the book's EM example).
- $\theta^{(i)}$, $z^{(i)}$: draws at iteration $i = 1, \ldots, 200$; the first 50 are burn-in.

Step — $p(z|\theta,\mathbf{y})$. Given $y_1 = x_1 + x_2 = 125$, each of the 125 outcomes falls in cell 2 with probability proportional to $\theta/4$ against $1/2$ for cell 1:

$$p_2 = \frac{\theta/4}{1/2 + \theta/4} = \frac{\theta}{2+\theta}, \tag{Q26.1}$$

$$Z \,|\, \theta, \mathbf{y} \sim \mathrm{Bin}\Big(125, \frac{\theta}{2+\theta}\Big). \tag{Q26.2}$$

Step — $p(\theta|\mathbf{y},z)$. With the flat prior, the posterior is the complete-data likelihood (C.3) with $x_2 = z$:

$$p(\theta|\mathbf{y},z) \propto \theta^{z+34}(1-\theta)^{18+20} = \theta^{z+34}(1-\theta)^{38}, \tag{Q26.3}$$

$$\theta \,|\, \mathbf{y}, z \sim \mathrm{Beta}(z+35,\ 39). \tag{Q26.4}$$

Step — algorithm and code (seed 5; function `ms_c5`). Start at $\theta^{(0)} = 0.5$ (the book's EM start) and alternate (Q26.2) and (Q26.4):

```
th = 0.5
for i in range(200):
    x2 = rng.binomial(125, th / (2 + th))     # z | theta, y        (Q26.2)
    th = rng.beta(x2 + 34 + 1, 18 + 20 + 1)   # theta | z, y        (Q26.4)
    out[i] = x2, th
keep = out[50:]                                # drop 50 burn-in values
```

First five θ draws: $0.5649, 0.5648, 0.5233, 0.6507, 0.6390$. Over the 150 retained draws: mean of θ 0.6155 (s.d. 0.0523), mean of Z 29.36.

Step — check against the exact posterior. Normalising (C.2) numerically gives $E(\theta|\mathbf{y}) = 0.6228$ and s.d. 0.0509; the posterior mode equals the MLE 0.62682 of Tabela C.1 (flat prior). The chain's mean is 0.0073 below the exact mean; with 150 draws the naive standard error is $0.0523/\sqrt{150} = 0.0043$, and positive autocorrelation inflates it, so the gap is ordinary Monte Carlo error. A longer run would shrink it.

Intuition. EM replaces the latent count by its expectation and maximises; the chained-data algorithm *draws* the latent count and then *draws* θ. Its draws describe the whole posterior (spread included), and its marginal for θ centres on the same value EM finds.

Ref: Morettin & Singer §C.2 (Exemplos C.2–C.3, pp. 480–482) and §B.7, Nota 2 (O Amostrador de Gibbs, p. 472); Greenberg §7.1 and §8.2 (latent data).